



Panipat Institute of Engineering and Technology
An Autonomous Institute Affiliated to Kurukshetra University, Kurukshetra

B. Tech. Computer Science Engineering (Semester-III)										
Scheme of Studies & Examination (w.e.f. session 2025-2026)										
Course No.	Course Title	Period (s)			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maxim um Marks	Maximum Marks	Maximum Marks	
Basic Science Course										
ASH-MAT-203A	Essential Mathematics for CSE	3	1	-	4	4	40	60	100	3 Hrs
Mandatory Course										
BT-CE-102A	Environmental Science	3	-	-	3	-	40	60	100	3 Hrs
Engineering Science Course										
BT-ECE-221A	Digital System Design	3	-	-	3	3	40	60	100	3 Hrs
BT-ECE-275A	Digital System Design Lab	-	-	4	4	2	50	50	100	3 Hrs
Professional Core Courses										
BT-CSE-201A	Database Management System	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-203A	Data Structures	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-205A	Object Oriented Programming	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-271A	Database Management System Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-273A	Data Structures Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-275A	Object Oriented Programming Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-277A	Summer Internship-I	-	-	-	-	2	50	50	100	3 Hrs
Total		18	1	16	35	26	490	610	1100	



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B. Tech. Computer Science Engineering (Semester-IV)										
Course No.	Course Title	Period (s)			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maximum Marks	Maximum Marks	Maximum Marks	
Engineering Science Courses										
BT- ECE-202A	Microprocessor based System Design	3	-	-	3	3	40	60	100	3 Hrs
BT- ECE-272A	Microprocessor based System Design	-	-	2	2	1	50	50	100	3 Hrs
Professional Core Courses										
BT-CSE-202A	Discrete Structures	3	1	-	4	4	40	60	100	3 Hrs
BT-CSE-204A	Computer Organization & Architecture	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-206A	Operating System	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-208A	Design & Analysis of Algorithms	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-276A	Operating Systems Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-278A	Design & Analysis of Algorithms Lab	-	-	4	4	2	50	50	100	3 Hrs
Total		15	1	8	24	20	300	400	700	
Honours Basket 1 for 4 th Semester (Computer Science & Engineering)										
BT-CSE-252A	Foundations of Machine Intelligence: Concepts, Techniques, and Applications	3	-	2	5	4	40	60	100	3 Hrs
BT-CSE-254A	Foundation of Data Science	3	-	2	5	4	40	60	100	3 Hrs
Exit Course after 4th Semester										
Course No.	Course Title	Teaching Schedule			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Max	Max	Max	
BT-CSE-305A	Software Engineering/ NPTEL/' Project	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-379A	Summer Internship-II	-	-	-	-	3	50	50	100	3 Hrs
Total (Exit Option)		3	0	0	3	6	90	110	200	
Grand Total (with Exit Option & without Honours)		36	2	24	62	52	880	1120	2000	
Note: 4 weeks Summer Internship-II is mandatory at the end of 4th semester (before the commencement of 5th semester).										
Students Exiting the program after Securing 95.5.0 credit will be awarded UG Diploma of computer Science Engineering.										
Students opting for the Honours program are required to choose any one of Two Courses & complete a minimum of 4 credits from designated honours Course Basket 1.										



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B. Tech. Computer Science Engineering (Semester-V)										
Course No.	Course Title	Period (s)			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maximum Marks	Maximum Marks	Maximum Marks	
Mandatory Credit Course										
ASH -HUM-210A	Indian Knowledge System: Concepts in Engineering	2	-	-	2	2	40	60	100	2 Hrs
Professional Core Courses										
BT-CSE-301A	Theory of Computation	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-303A	Fundamentals of Artificial	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-305A	Software Engineering	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-307A	Fundamentals of Machine Learning	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE- CS-309A	Fundamentals of Cyber Security	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-375A	Software Engineering Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-377A	Machine Learning Lab	-		2	2	1	50	50	100	3 Hrs
BT-CSE-379A	Summer Internship-II	-	-	-	-	2	50	50	100	3 Hrs
Total		17	0	6	23	22	390	510	900	
Honours Basket 1 for 5 th Semester (Computer Science & Engineering)										
BT-CSE-351A	Fundamentals of Deep Learning	3	-	2	5	4	40	60	100	3 Hrs
BT-CSE-353A	Data Warehousing and Mining	3	-	2	5	4	40	60	100	3 Hrs
Students opting for the Honours program are required to choose any one of Two Courses & complete a minimum of 4 credits from designated honours Course Basket 2.										



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B. Tech. Computer Science Engineering (Semester-VI)										
Course No.	Course Title	Period (s)			Hours/Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maximum Marks	Maximum Marks	Maximum Marks	
Humanities										
MH-902A	Business Intelligence & Entrepreneurship	3	-	-	3	3	40	60	100	3 Hrs
Professional Core Courses										
BT-CSE-302A	Compiler Design	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-304A	Computer Networks	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE3***A	Open Elective-I	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE***A	Program Elective-I	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-370A	Project-1	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-374A	Computer Networks Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE***-A	Elective-I Lab	-	-	4	4	2	50	50	100	3 Hrs
Total		15	0	12	27	21	350	450	800	
OE Elective-I		PEC Elective-I					PEC Elective LAB-I			
Advanced Computer Architecture: BT-CSE-352A		Fundamentals of Data Science: BT-CSE-358A					Data Science Lab: BT-CSE-378A			
Distributed Systems: BT-CSE-354A		Unix & Linux Programming: BT-CSE-360A					Unix & Linux Programming Lab: BT-CSE-370A			
Information Security & Cryptography: BT-CSE-356A		Game Development Fundamental using Blender: BT-CSE-362A					Game Development Fundamental using Blender Lab: BT-CSE-372A			
Honours Basket 3 for 6 th Semester (Computer Science & Engineering)										
BT-CSE-364A	Soft Computing	3	-	2	5	4	40	60	100	
BT-CSE-366A	Data Engineering	3	-	2	5	4	40	60	100	
Exit Course after 6th Semester										
Course No.	Course Title	Teaching Schedule			Hours/Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Max	Max	Max	
BT-CSE-401A	Neural Network & Deep Learning/ NPTEL/ Project	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-473A	Summer Internship -III	-		-	3	3	50	50	100	3 Hrs
Total (Exit Option)		3	0	0	6	6	90	110	200	3 Hrs
Grand Total (with Exit Option & without Honours)		35	0	18	56	49	830	1070	1900	
Note: 6 weeks Summer Internship-III is mandatory at the end of 6th semester (before the commencement of 7th semester).										
Students Exiting the program after Securing 139.5 credit will be awarded U.G Degree of computer Science Engineering.										
Students opting for the Honours program are required to choose any one of Two Courses & complete a minimum of 4 credits from designated honours Course Basket 3.										



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Course No.	Course Title	Period (s)			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maximum Marks	Maximum Marks	Maximum Marks	
Professional Core Courses										
BT-CSE-401A	Neural Networks and Deep Learning	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE***A	Program Elective-II	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE***A	Open Elective-II	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-481A	Project-II	-	-	6	6	3	50	50	100	3 Hrs
BT-CSE***A	Program Elective-II Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-471A	Neural Networks and Deep Learning Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-473A	Summer Internship-III	-	-	-	-	2	50	50	100	3 Hrs
Total		9		14	23	18	320	380	700	
Honours Basket 4 for 7 th Semester (Computer Science & Engineering)										
BT-CSE-463A	Generative Artificial Intelligence	3	-	2	5	4	40	60	100	3 Hrs
BT-CSE-465A	Data Analytics with Visualization	3	-	2	5	4	40	60	100	3 Hrs
Students opting for the Honours program are required to choose any one of Two Courses & complete a minimum of 4 credits from designated honours Course Basket 4.										
PE Elective-II		PE Elective-II-LAB					OE Elective-II			
Software Verification and Validation and Testing: BT-CSE-455A		Software Verification and Validation and Testing Lab: BT-CSE-475A					Advance Data Base Management System: BT-CSE-451A			
Information Retrieval: BT-CSE-457A		Information Retrieval Lab: BT-CSE-477A					Quantum Computing: BT-CSE-453A			
Game Development using Unity: BT-CSE-459A		Game Development using Unity Lab: BT-CSE-479A					Natural Language Processing: BT-CSE461A			



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B. Tech. Computer Science Engineering (Semester-VIII)										
Course No.	Course Title	Period (s)			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maximum Marks	Maximum Marks	Maximum Marks	
Professional Core Courses										
BT-CSE-402A	Full Stack Development	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE***A	Elective-III	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE***A	Open Elective-III	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-472A	Full Stack development Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-474A	Project-III	-	-	6	6	3	60	40	100	3 Hrs
BT-CSE-476A	Seminar			2	2	1	50	-	50	3 Hrs
Total		9	0	12	21	15	270	280	550	
Honours Basket 5 for 8 th Semester (Computer Science & Engineering)										
BT-CSE-464A	Computer Vision	3	-	2	5	4	40	60	100	3 Hrs
BT-CSE-466A	Business Intelligence	3	-	2	5	4	40	60	100	3 Hrs
Grand Total without Honours		18	0	26	44	33	590	660	1250	
Note: Students opting for the Honors program must choose one of the two courses offered in the Honors Basket-5 to earn 4 credits. In order to qualify for the B.Tech. (CSE) with Honors degree, they must complete a total of 20 credits from the designated Honors Course Baskets.										
To be eligible for the award of the B.Tech. in Computer Science & Engineering, students must successfully complete a minimum of 166.5 credits as specified in the program curriculum.										
To be awarded the B.Tech. (CSE) with Honors degree, students must complete a total of 186.5 credits-comprising the standard 166.5 credits plus an additional 20 honors credits drawn from the approved Honors course basket.										
Program Elective-III						Open Elective-III				
Cloud Computing BT-CSE-452A						Mobile Application Development BT-CSE-458A				
Human Computer Interaction BT-CSE-454A						Blockchain BT-CSE-460A				
Software Project Management BT-CSE-456A						IOT Architectures and Protocols BT-CSE-462				



Minor Scheme

Department of Computer Science & Engineering offering Minor Degree in Web Development to other departments										
Semester	Course No.	Course Title	Period(s)			Credit (s)	CIE	SEE	Total	Duration
			L	T	P		Max	Max		
4th Sem	BT-CSE-212A	Foundations of Web Design and Development	3			3	40	60	100	3 Hrs
	BT-CSE-272A	Foundations of Web Design and Development Lab			2	1	50	50	100	3 Hrs
5th Sem	BT-CSE-311A	Database Technologies for Web Applications	3			3	40	60	100	3 Hrs
	BT-CSE-371A	Database Technologies for Web Applications Lab			2	1	50	50	100	3 Hrs
6th Sem	BT-CSE-306A	Dynamic web Application using PHP	3			3	40	60	100	3 Hrs
	BT-CSE-380A	Dynamic Web Application using PHP lab			2	1	50	50	100	3 Hrs
7th Sem	BT-CSE-403A	Advanced Web Frameworks	3			3	40	60	100	3 Hrs
	BT-CSE-481A	Minor project-I			4	2	50	50	100	3 Hrs
8th Sem	BT-CSE-478A	Project-II			6	3	50	50	100	3 Hrs
Total			12		16	20	410	490	900	



Syllabus

3rd Semester



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Course No.	Course Title	Period (s)			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maxim um Marks	Maximum Marks	Maximum Marks	
Basic Science Course										
ASH-MAT-203A	Essential Mathematics for CSE	3	1	-	4	4	40	60	100	3 Hrs
Mandatory Course										
BT-CE-102A	Environmental Science	3	-	-	3	-	40	60	100	3 Hrs
Engineering Science Course										
BT-ECE-221A	Digital System Design	3	-	-	3	3	40	60	100	3 Hrs
BT-ECE-275A	Digital System Design Lab	-	-	4	4	2	50	50	100	3 Hrs
Professional Core Courses										
BT-CSE-201A	Database Management System	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-203A	Data Structures	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-205A	Object Oriented Programming	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-271A	Database Management System Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-273A	Data Structures Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-275A	Object Oriented Programming Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-277A	Summer Internship-I	-	-	-	-	2	50	50	100	3 Hrs
Total		18	1	16	35	26	490	610	1100	



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Course Code				Course Title						L	T	P	Credits	
ASH-MAT-203A				Essential Mathematics for CSE						3	1	0	4	
										CIE	SEE		Total	
										40	60		100	
Course Outcomes														
CO1	Describe the concept of Laplace transform and how it is useful in solving the definite integrals and initial value problems.													
CO2	Solve the Partial Differential Equations for multivariable differential equations originated from real world problems.													
CO3	Solve the problems using numerical methods in a comprehensive manner and describe the essential tool of Numerical Integration needed in approximate solutions													
CO4	Apply the Concepts of functions of complex variable and conformal mapping to solve complex integral and analyze analytic function.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2:														
COs\POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	-	-	-	-	-	-	1	3	2
CO2	3	3	3	3	2	-	-	-	-	-	-	2	3	3
CO3	3	3	3	3	3	-	-	-	-	-	-	2	3	3
CO4	3	2	2	2	2	-	-	-	-	-	-	1	3	2
Avg	3	2.5	2.5	2.5	2.25							1.5	3	2.5

Course Outline

Unit-1 Laplace Transformation

Contact Hours: 12

Definition, Laplace transformation of basic functions, existence condition for Laplace transformation, Properties of Laplace transformation (Linearity, scaling and shifting). Unit step function, Impulse Function, Periodic Function. Laplace transformation of derivatives, Laplace transformation of integrals, Differentiation of transforms, integration of transforms, Convolution theorem, inverse Laplace Transformation. Solution of ordinary differential equation,

Unit-2 Partial differential Equations

Contact Hours: 8

Formation of partial differential equation, Linear Equation of First Order, Non-Linear Equation of First Order, Charpits Method, Method of Separation of Variables. Application of partial differential equation to solve heat equation and wave equation.

Unit-3: Numerical Methods

Contact Hours: 11

Interpolation: Finite differences operators, Interpolation using Newton's forward formula and Backward Difference formula, Interpolation with unequal intervals: Lagrange's formulae.
 Numerical Integration: Trapezoidal rule, Simpson's 1/3rd rule, Simpson's 3/8 rule, Ordinary differential equations: Runge Kutta 4th order method of solving first and second order equations

Unit-4: Functions of Complex Variables

Contact Hours: 10

Functions of a complex variable, Exponential function, Trigonometric, Hyperbolic and Logarithmic functions, Differentiability and analyticity. Cauchy-Riemann equations, Necessary and sufficient conditions for a function to be analytic, Harmonic functions, Complex integrals, Cauchy theorem, Cauchy Integral Formula, Classification of Singularities, Residues, Cauchy Residue theorem.



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Text Books:

1. N.P. Bali and Manish Goyal , A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2010.
2. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 35th Edition, 2000.

Other References:

1. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
2. Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2008.
3. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
4. W. E. Boyce and R. C. DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th Edition, Wiley India, 2009

NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions is to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



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Course Code	Course Title	L	T	P	Credits
BT-CE-102A	Environmental Studies	3	0	0	3
		CIE	SEE		Total
		40	60		100

Course Outcomes

CO1	The students will be able to understand the importance of natural resources.
CO2	To learn the theoretical concepts of ecosystem and biodiversity conservation.
CO3	Students will be able to understand about pollution and its control.
CO4	The students will be able to understand the basic concept of sustainable development.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium,

COs\POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	2	3	-	-	-	-	2	2	-
CO2	3	2	2	3	-	-	3	-	-	-	-	2	2	1
CO3	2	3	2	2	-	3	3	-	-	-	-	1	1	-
CO4	2	1	2	-	-	-	3	-	-	-	-	3	1	2

Course Outline

Unit –1: The multidisciplinary nature of environmental studies

Contact Hours:10

Definition, Scope, and Importance. Need for public awareness. Natural Resources: Renewable and NonRenewable Resources: Natural resources and associated problems. (a) Forest Resources: Use and overexploitation, deforestation, case studies. (b) Energy Resources- Growing energy needs, renewable and nonrenewable energy sources, use of alternate energy sources, case studies. (c) Land Resources- Land as a resource, land degradation, man-induced landslides, soil erosion, and desertification. Role of an individual in the conservation of natural resources. Equitable use of resources for a sustainable lifestyle.

Unit – 2: Ecosystem and Biodiversity Concept

Contact Hours:10

Ecosystem: Concept of an ecosystem, Structure and function of an ecosystem, Producers, consumers and decomposers, Energy flow in the ecosystem. Food Chains, food webs. Biodiversity and its conservation: Introduction, Definition: genetic, species and ecosystem diversity. Bio- geographical classification of India, Value of biodiversity, Hot spots of Biodiversity. Threats to biodiversity: Habitat loss, poaching of wildlife, man-wildlife conflicts. Endangered and endemic species of India, Conservation of Biodiversity- In situ and Ex-Situ biodiversity conservation.

Unit -3: Environmental Pollution

Contact Hours:10

Definition, Cause, effects, and control measures of- (a) Air Pollution, (b) Water Pollution, (c) Soil Pollution, (d) Marine Pollution. Solid waste management- cause, effects, and control measures of urban and industrial wastes. Role of an individual in preventing pollution, E-waste management, and plastic waste. Disaster management- Natural: floods, earthquakes, cyclones, and landslides, etc. Man-made Disaster: Fire, Industrial pollution, nuclear disaster, biological Disaster, Disaster preparedness plans.

Unit – 4: Social Issues and the Environment

Contact Hours:11



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From unsustainable to sustainable development, Urban problems related to energy, Water conservation, rainwater harvesting, and watershed management. Resettlement and rehabilitation of people: Its problems and concerns. Environmental ethics issues and possible solutions, Climate change, global warming, acid rain, ozone layer depletion, Case studies, Concept of green building, carbon footprinting. Environment Protection Act (Air, water, wildlife, forest), Environment and human health, Role of Information Technology in Environment and Human Health, Case Studies. Emerging technologies for Environmental management.

Text Book

1. Environmental Studies- Deswal and Deswal. Dhanpat Rai & Co.
2. Environmental Science & Engineering Anandan, P. and Kumaravelan, R. 2009. Scitech Publications (India) Pvt. Ltd., India

Reference Books

1. Environmental Studies. Daniels Ranjit R. J. and Krishnaswamy. 2013. Wiley India.
2. Environmental Science- Botkin and Keller. 2012. Wiley, India.

NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions is to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



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Course Code	Course Title	L	T	P	Credits
BT-ECE-221A	Digital System Design	3	0	0	3
		CIE	SEE		Total
		40	60		100

Course Outcomes

CO1	Summarize the number systems and understand the basic characteristics of logic families.
CO2	Illustrate the various Boolean postulates and Minimization techniques.
CO3	Design and analysis of different type of combinational circuits.
CO4	Design and analysis of synchronous and asynchronous sequential circuits using flip flop.
CO5	Illustrate the working of various counters in digital system design.
CO6	Illustrate the working of different A/D and D/A converters.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

Co / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3				-	-	-	-	-	-	-	-	-	-
CO2	3	3			-	-	-	-	-	-	-	-	-	-
CO3	3		3		-	-	-	-	-	-	-	-	-	-
CO4	3	3	3	2	-	-	-	-	-	-	-	-	-	-
CO5	3				-	-	-	-	-	-	-	-	-	-
CO6	3		3	2	-	-	-	-	-	-	-	-	-	-
Avg	3	3	3	2	-	-	-	-	-	-	-	-	-	-

Course Outline

Unit-1: Fundamentals of Digital Systems

Contact Hours: 12

Fundamentals of Digital Systems: Logic Gates, Universal Gates, Number systems: binary, signed binary, octal, hexadecimal number, binary arithmetic, one's and two's complements arithmetic, Minimization Techniques: Boolean Algebra, Standard representation of logic functions: SOP and POS forms, Simplification of switching functions using K-map and Quine-McCluskey tabular methods.

Unit-2: Combinational Circuits

Contact Hours: 10

Combinational Digital Circuits: Design procedure: Half adder, Full Adder, Half subtractor, Full subtractor, Parallel binary adder, parallel binary Subtractor, BCD adder, Multiplexer and Demultiplexer, decoder, encoder, parity checker, parity generators, Magnitude Comparator.

Unit-3: Sequential Circuits

Contact Hours: 10

Sequential circuits: A 1-bit memory, Latch, the clocked SR flip flop, J- K, T and D types flip flops, Interconversion of flipflops, applications of flip flops, shift registers, Bi-directional Shift Register, Synchronous and Asynchronous counter, Mod -N counter.

Unit-4: Converters and Memory Devices

Contact Hours: 12

A/D and D/A Converters: Digital to analog converters: weighted resistor/converter, R-2R Ladder D/A converter, specifications for D/A converters, analog to digital converters: quantization and encoding, A/D converter, successive approximation A/D converter, Dual slope A/D converter specifications for A/D converters.



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Classification of memories: ROM, RAM, PROM, EPROM, EEPROM, EAPROM, Programmable Logic Devices - Programmable Logic Array (PLA), Programmable Array Logic (PAL), Implementation of PLA, PAL using ROM.

Text Books:

1. Donald P. Leach and Albert Paul Malvino, Digital Principles and Applications, 8th Edition, TMH, 2003.M.
2. Morris Mano, Digital Design, 3rd Edition, Prentice Hall of India Pvt. Ltd., 2003 / Pearson Education (Singapore) Pvt. Ltd., New Delhi, 2003.
3. A. A. Kumar: Fundamentals of digital circuits, Prentice Hall of India.

Reference Books:

1. A.K. Maini, Digital Electronics, Wiley India
2. John F. Wakerly, Digital Design, Fourth Edition, Pearson/PHI, 2006
3. John. M Yarbrough, Digital Logic Applications and Design, Thomson Learning, 2002.
4. S. Salivahanan and S. Arivazhagan, Digital Circuits and Design, 3rd Edition., Vikas Publishing House Pvt. Ltd, New Delhi, 2006.

NOTE:

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Course Code		Course Title								L	T	P	Credits		
BT-ECE-275A		Digital System Design Lab								--	0	4	2		
										CIE		SEE		Total	
										50		50		100	
Pre-requisites: Basic Electronics Engineering															
Course Outcomes															
CO1	Identify various digital ICs and understand their operation														
CO2	Design basic combinational Circuits and verify their functionalities														
CO3	Apply the design procedures to design basic sequential circuits using Flip-Flops														
CO4	Understand the concept of encoders and decoders and verify their operation														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
Co /	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
CO1	3	-	-	-	-	-	-	2	2	1		1	3	-	
CO2	3	2	3	1	-	-	-	2	2	1		1	3	2	
CO3	3	2	3	1	-	-	-	2	2	1		1	3	2	
CO4	3	2	3	1	-	-	-	2	2	1		1	3	2	
Avg	3	2	3	1	-	-	-	2	2	1		1	3	2	

List of experiments:

1. Familiarization with Digital Trainer Kit and associated equipment.
2. Study of TTL gates AND, OR, NOT, NAND, NOR, EX-OR, EX-NOR.
3. Design and realize a given function using K-Maps and verify its performance.
4. To verify the operation of Multiplexer and De-multiplexer.
5. To verify the operation of Comparator.
6. To verify the truth table of S-R, J-K, T, D Flip-flops.
7. To verify the operation of Bi-directional shift register.
8. To design and verify the operation of 3-bit asynchronous counter.
9. To design and verify the operation of asynchronous Up/down counter.
10. To design and verify the operation of asynchronous Decade counter.
11. Study of Encoder and Decoder.
12. Study of BCD to 7 segment Decoder

NOTE: A student has to perform at least ten experiments. Seven experiments should be performed from the above list. Remaining three experiments may either be performed from the above list or as per the scope of the syllabus



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Course Code				Course Title						L	T	P	Credits	
BT-CSE-201E				Database Management Systems						3	-	-	3	
										CIE	SEE	Total		
										40	60	100		
Pre-requisites: Professional Core Course														
Course Outcomes														
CO1	Understand the basic concepts of databases, data models, ER diagrams, and database architecture to model and design structured data storage.													
CO2	Apply relational algebra, relational calculus, and SQL to create, retrieve, and manipulate data in relational databases.													
CO3	Analyze functional dependencies and apply normalization techniques to design efficient and consistent database schemas.													
CO4	Demonstrate knowledge of transaction processing, concurrency control, database recovery, and security features for reliable database operations.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
COs\POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	-	-	-	-	-	-	-	2	1
CO2	2	3	2	-	-	-	-	-	-	-	-	-	3	1
CO3	1	2	3	1	-	-	-	-	-	-	-	-	2	2
CO4	-	2	3	2	1	-	-	-	-	-	-	-	1	3
Avg	2	2.25	2.3	1.5	1	-	-	-	-	-	-	-	2	1.75

Course Outline:

Unit-1: Fundamentals of Databases and Data Modeling

Contact Hours: 11

Data, Information, Records, Files, Schema and Instance, Advantages of DBMS over file processing systems, Responsibilities of Database Administrator, Three-level architecture of database systems, Data Abstraction, Data Independence, Hierarchical Data Model, Network Data Model, Relational Data Model, Entity-Relationship Model, Types of Attributes, Keys, Integrity Constraints, Domain and Tuple Constraints, Designing of ER Diagram, Design Issues, Reduction of ER diagram into Relational Tables

Unit-2: Relational Model and Query Languages

Contact Hours: 10

Introduction to Query Languages, Relational Algebra: Select, Project, Join, Union, Intersection, Difference, Cartesian Product, Relational Calculus: Tuple Relational Calculus, Domain Relational Calculus, SQL Components: Data Definition Language (DDL), Data Manipulation Language (DML), Data Control Language (DCL), SQL Queries using WHERE, GROUP BY, HAVING, ORDER BY clauses, Nested Queries and Subqueries, Views, Indexes, Constraints (NOT NULL, UNIQUE, CHECK, DEFAULT, PRIMARY KEY, FOREIGN KEY).

Unit-3: Normalization and File Organization:

Contact Hours: 10

SQL Functional Dependency, Need and Benefits of Normalization, 1NF, 2NF, 3NF, BCNF, Multivalued Dependencies, 4NF, Join Dependencies, 5NF, Sequential Files, Index Sequential Files, Direct Files, Hashing, B-trees Index Files, Inverted Lists.



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Unit-4 : Transactions, Security, and Advanced Topics

Contact Hours: 10

ACID Properties, Operations on Transactions, Serializability of Scheduling, Concurrency Problems, Concurrency Control Techniques, Locking Schemes, Deadlock and Methods for Handling Deadlock, Database Backup and Recovery, Recovery Techniques, Shadow Paging, Database Security, Authentication, Authorization, Access Control Models: DAC, MAC, RBAC, Intrusion Detection, SQL Injection, Introduction to Distributed Data Processing, Data Mining, Data Warehousing.

Text Books:

1. “Database System Concepts”, 6th Edition by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill.
2. Introduction to Database Management system by Bipin Desai, 1991, Galgotia Pub.

Reference Books:

1. “Principles of Database and Knowledge–Base Systems”, Vol1 by J. D. Ullman, Computer Science Press.
2. “Fundamentals of Database Systems”, 5th Edition by R. Elmasri and S. Navathe, Pearson Education.
3. “Foundations of Databases”, Reprint by Serge Abiteboul, Richard Hull, Victor Vianu, Addison-Wesley.

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Course Code				Course Title							L	T	P	Credits
BT-CSE-203A				Data Structures							3	-	-	3
											CIE	SEE	Total	
											40	60	100	
Course Outcomes														
CO1	To understand and apply basic data structures such as arrays, and implement searching and sorting algorithms while analyzing their time and space complexity.													
CO2	To apply insertion, deletion, and traversal techniques on data structures.													
CO3	To create and manipulate dynamic data structures using various types of linked lists and implement stack and queue operations using linked list representations.													
CO4	To implement and traverse tree and graph data structures, and apply graph algorithms for solving computational problems.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
Cos/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	-	1	-	2	3	-
CO2	3	2	-	-	1	1	-	-	-	1	-	2	3	-
CO3	3	2	2	2	1	1	-	-	-	1	1	2	2	2
CO4	3	3	2	2	1	1	-	-	-	1	-	2	2	2
AVG	3	2.25	2	2	1	1	-	-	-	1	1	2.00	2.66	2

Course Outline

Unit 1: Fundamentals of Data Structures and Algorithms

Contact Hours 12

Definition and need of data structures, classification of data structures (linear and non-linear), abstract data types (ADT), primitive and non-primitive data types, built-in and user-defined data types, algorithm basics, time and space complexity, Big O, Omega (Ω), and Theta (Θ) notations, best, worst, and average case analysis.

Arrays: one-dimensional arrays (declaration, initialization, memory representation), array operations (traversal, insertion, deletion), two-dimensional arrays (concept and applications), linear search, binary search, bubble sort, selection sort, insertion sort, introduction to recursion, examples of recursion (factorial, Fibonacci).

Unit-2: Stacks and Queues

Contact Hours 12

Stack: Definition and applications of stacks, array-based implementation of stacks, stack operations (push, pop, peek), evaluation of infix, prefix, and postfix expressions, conversion between infix, prefix, and postfix expressions, introduction to merge sort, introduction to quick sort.

Queues: Definition and applications of queues, array-based implementation of linear queues, queue operations (enqueue, dequeue), circular queue and its implementation, priority queue basics and applications.

Unit-3: Linked Lists:

Contact Hours 8

Need for dynamic data structures, definition and applications of linked lists, singly linked list (creation, traversal, insertion, deletion), comparison between static and dynamic memory allocation, doubly linked list (creation, insertion, deletion), circular linked list (creation, traversal), implementation of stack and queue using linked lists.



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Unit-4: Trees and Graph

Contact Hours 10

Definition and terminology of trees, binary tree and its applications, static and dynamic implementation of binary trees, binary tree traversals (preorder, inorder, postorder), expression representation using trees (infix, prefix, postfix), introduction to binary search trees, introduction to heaps and heap sort,

Graphs: Definition and terminology of graphs, types of graphs (directed, undirected), memory representation of graphs (adjacency matrix and list), graph traversal techniques (BFS, DFS), introduction to minimum spanning trees.

Suggested Books:

1. Theory and Problems of Data Structures by Jr. Seymour Lipschetz, Schaum's outline, TMH.
2. Data Structures and Algorithms by PAI, TMH.
3. Fundamentals of Data structures by Ellis Horowitz and Sartaj Sahni, Pub, 1983, AW.
4. Data Structures and Algorithms by A.V. Aho, J.E. Hopcroft and T.D. Ullman, Original edition, Addison-Wesley, 1999, Low Priced Edition.

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Course Code				Course Title						L	T	P	Credits	
BT-CSE-205A				OBJECT ORIENTED PROGRAMMING						3	-	-	3	
										CIE	SEE	Total		
										40	60	100		
Purpose														
To equip students with a strong foundation in java programming and object-oriented principles, enabling them to design, develop, and implement efficient, reusable, and secure software solutions														
Course Outcomes														
CO1		Understand the architecture of Java and the Java Virtual Machine (JVM), and apply object-oriented concepts.												
CO2		Develop modular Java applications by applying packages, access modifiers, operators, and control flow statements.												
CO3		Implement object-oriented design principles such as inheritance, polymorphism, interfaces, and method overriding to build maintainable and extensible Java applications.												
CO4		Create multithreaded and GUI-based applications using Java concurrency constructs and JavaFX event-handling mechanisms.												
Course Outcomes (CO) to Programme Outcomes (PO) Mapping (Scale 1: Low, 2: Medium, 3: High)														
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2									2	1	2
CO2	3	2	3		2							2	2	2
CO3	3	3	3		3					2		2	2	2
CO4	3	2	3		3					2		2	2	2
AVG	3	2	2.5		2.6		-	-		2		2	2	2

Course Outline

Unit-1: Introduction

Contact Hours 12

The overview of Java's architecture and the architecture of the Java Virtual Machine (JVM). Classes: Declaring Members (Fields and Methods), Instance Members, Static Members. Objects: Class Instantiation, Reference Values, and References, Object Aliases. Basic Language Elements, Primitive Data Types, Variable Declarations, Initial Values for Variables, Class Declarations, Method Declarations, this reference, Method Overloading, Constructors, The Default Constructor and Constructors overloading. Arrays, Anonymous Arrays, Multidimensional Arrays, Variable Arity Methods, The main() Method, Program Arguments

Unit-2: Packages:

Contact Hours 12

Defining Packages, Using Packages, Compiling Code into Packages, Running Code from Packages. Scope Rules, Accessibility Modifiers, Overview of other Modifiers for Members. Operators and Expressions, Overview of Control Flow Statements. Exception Handling: The try Block, The catch Block, The finally Block, The throw Statement, The throws Clause, Checked and Unchecked Exceptions, Defining New Exceptions

Unit-3: Object-Oriented Programming

Contact Hours 12

Object-Oriented Programming: Single Implementation Inheritance, Overriding Methods, Hiding Members, The Object Reference super, Chaining Constructors Using this() and super() Interfaces: Defining Interfaces, Abstract Method Declarations, Implementing Interfaces, Extending Interfaces, Interface References, Constants in Interfaces, Polymorphism and Dynamic Method Lookup. Fundamental Classes: Overview of the java.lang Package, The Object Class, The Wrapper Classes, The String Class, The StringBuilder and the String Buffer Classes.



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Unit-4: Multithreading:

Contact Hours 8

Overview of Threads, the Main Thread, Thread Creation, Synchronization, Thread Transitions. Basics of Event Handling, Graphics Programming using Java Fx.

Reference Books:

1. Y. Daniel Liang: Introduction to Java Programming, Comprehensive Version, Pearson Education, 12th Ed.
2. Bruce Eckel, Thinking In Java, Pearson Education, 4th Ed., 2006.
3. Dietel & Deitel, Java How to Program, Pearson Education, 10th Ed., 2015.
4. Kathy Sierra & Bert Bates, Head First Java, O'REILLY, 2nd Ed., 2005.
5. Cay s. Horstmann & Gary Cornell, Core Java. Volume I, Fundamentals, Sun Microsystems Press, 8 th Ed., 2008.

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Panipat Institute of Engineering and Technology
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Course Code					Course Title						L	T	P	Credits
BT-CSE-271A					Database Management Systems lab						-	-	4	2
											CIE	SEE	Total	
											50	50	100	
Course Outcomes														
CO1		Understanding SQL and Database Installation.												
CO2		Defining and Manipulating Database Structures Using SQL Commands.												
CO3		Advanced Querying and Data Analysis.												
CO4		Implementing Advanced Database Features.												
Course Outcomes (CO) to Programme Outcomes (PO) Mapping (Scale 1: Low, 2: Medium, 3: High)														
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	2	-	-	-	-	-	-	2	2	2
CO2	3	2	2		2	-	-	-	-	1	-	2	2	2
CO3	3	3	2	2	2	-	-	-	-	2	-	2	2	2
CO4	3	2	3	2	3	-	-	-	-	2	-	2	2	2
AVG	3	2	3	2	2.25	-	-	-	-	1.6	-	2	2	2

List of Experiments:

1. Introduction to SQL and installation of SQL Server / Oracle.
2. Data Types, Creating Tables, Retrieval of Rows using Select Statement, Conditional Retrieval of Rows, Alter and Drop Statements.
3. Working with Null Values, matching a Pattern from a Table, Ordering the Result of a Query, Aggregate Functions, Update and Delete Statements.
4. Create a database and write the programs to carry out the following operation:
 - (a) Add a record in the database
 - (b) Delete a record in the database
 - (c) Modify the record in the database
 - (d) Generate queries, e) Data operations
5. Create a view to display details of employees working on more than one project.
6. Create a view to display details of employees not working on any project.
7. Write an experiment to perform all JOIN operation.
8. Write an Experiment to Implement SQL commands that manage user permissions and control transactions i.e. GRANT, REVOKE, COMMIT, and ROLLBACK
9. Write trigger for before and after insertion. Detection and updation process.
10. Write a procedure to give incentive to employees working on all projects. If no such employee found give app.
11. Write a procedure for computing amount telephone bill on the basic of following conditions.

NOTE:

A student has to perform at least ten experiments. Seven experiments should be performed from the above list. Remaining three experiments may either be performed from the above list or as per the scope of the syllabus.



Panipat Institute of Engineering and Technology
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Course Code				Course Title							L	T	P	Credits	
BT-CSE-273A				Data Structures Lab							-	-	4	2	
											CIE	SEE	Total		
											50	50	100		
Course Outcomes															
CO1		To introduce the basic concepts of Data structure, basic data types, searching and sorting based on array data types.													
CO2		To introduce the structured data types like Stacks and Queue and its basic operation’s implementation.													
CO3		To introduce dynamic implementation of linked list.													
CO4		To introduce the concepts of Tree and graph and implementation of traversal algorithms.													
Course Outcomes (CO) to Programme Outcomes (PO) Mapping (Scale 1: Low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
CO1	3	3	3	2	2	1	-	1	1	-	-	2	2	2	
CO2	3	3	3	3	2	1	-	1	2	1	-	3	2	2	
CO3	3	3	3	3	2	1	-	1	2	1	1	3	2	2	
CO4	3	3	3	2	2	1	-	1	2	1	1	2	2	2	
AVG	3	3	3	2.50	2.00	1	-	1	1.75	1	1.00	2.5	2	2	

List of Experiments:

1. Write a program for Binary search methods.
2. Write a program for insertion sort, selection sort and bubble sort.
3. Write a program to implement Stack and its operation.
4. Write a program for quick sort.
5. Write a program for merge sort.
6. Write a program to implement Queue and its operation.
7. Write a program to implement Circular Queue and its operation.
8. Write a program to implement singly linked list for the following operations: Create, Display, searching, traversing and deletion.
9. Write a program to implement doubly linked list for the following operations: Create, Display, inserting, counting, searching, traversing and deletion.
10. Write a program to implement circular linked list for the following operations: Create, Display, inserting, counting, searching, traversing and deletion.

NOTE:

A student has to perform at least ten experiments. Seven experiments should be performed from the above list. Remaining three experiments may either be performed from the above list or as per the scope of the syllabus



Panipat Institute of Engineering and Technology
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Course Code				Course Title							L	T	P	Credits	
BT-CSE-275A				OBJECT ORIENTED PROGRAMMING LAB							-	-	4	2	
											CIE	SEE	Total		
											50	50	100		
Purpose															
To develop hands-on skills in applying object-oriented programming concepts using Java for designing modular, reusable, and efficient software solutions															
Course Outcomes															
CO1	Apply the basic concepts of Java programming such as data types, control statements, arrays, classes, objects, constructors, and methods to solve computational problems.														
CO2	Demonstrate object-oriented programming concepts including inheritance, polymorphism, abstraction, apsulation, interfaces, exception handling, and multithreading through Java applications.														
CO3	Implement advanced Java features such as packages, wrapper classes, collections framework.														
CO4	Develop real-time applications using Java by integrating GUI programming.														
Course Outcomes (CO) to Programme Outcomes (PO) Mapping (Scale 1: Low, 2: Medium, 3: High)															
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
CO1	3	3	2	1	2	-	-	-	-	-	-	2	2	2	
CO2	2	3	3	2	2	-	-	1	1	1	-	2	2	2	
CO3	3	2	3	2	3	-	-	1	2	2	2	3	2	2	
CO4	2	3	3	2	3	1	-	1	3	2	2	3	2	2	
AVG	2.5	2.75	2.75	1.75	2.75	1	-	1	1.75	1.25	2	2.5	2	2	

List of Experiments:

1. Develop a Java program to demonstrate class creation, object instantiation, use of instance variables, and static members.
2. Implement a Java program to declare and initialize primitive data types, utilize the this keyword, and demonstrate method overloading.
3. Create a Java class that implements both default and parameterized constructors, including constructor overloading
4. Write a Java program to generate the Fibonacci series using both recursive and non-recursive approaches.
5. Develop a Java application to perform matrix multiplication using two-dimensional arrays.
6. Create a Java program to accept and display employee details using the Scanner class.
7. Implement a program that demonstrates the use of single-dimensional, multi-dimensional, and anonymous arrays.
8. Write a Java program to demonstrate the use of control flow statements (if, switch, for, while, do-while) and arithmetic/logical operators.
9. Create a user-defined package in Java and demonstrate importing and accessing its classes from another program.
10. Implement a program using exception handling constructs including try, catch, finally, throw, and throws for both checked and unchecked exceptions.
11. Write a Java program to define a custom exception by extending the Exception class and demonstrate its usage.



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12. Develop a Java program that demonstrates multiple catch blocks and the sequence in which exceptions are handled.
13. Implement a Java program to check whether a given string is a palindrome using the StringBuilder class.
14. Write a program to demonstrate single inheritance, method overriding, and usage of the super keyword.
15. A) Develop a Java program to implement an abstract class and its subclass.
B) Create a Java program to implement interfaces and demonstrate polymorphism using interface references.
16. A) Write a program to create and access inner classes to demonstrate encapsulation.
B) Develop a program to extend and implement multiple interfaces in Java.
17. Write a program to demonstrate the usage of wrapper classes and override methods from the Object class such as to String(), equals(), and hash Code().
18. Create a Java program that implements a multi-threaded application with three concurrently running threads.
19. Solve the classic producer–consumer problem using threads and synchronization in Java.
20. Build a simple GUI application using JavaFX that handles basic user events such as button clicks or input actions.
21. Write a Java program to demonstrate file handling by displaying file properties and reading/writing text data using the File class and streams.
22. A) Create a program using Array List to store and display a list of student names.
B) Write a program to load name and phone number pairs from a text file into a Hash table and retrieve data.
23. Develop a Java program that connects to a relational database using JDBC and performs basic operations such as insert and retrieve.

NOTE:

A student has to perform at least Fifteen experiments. Ten experiments should be performed from the above list. Remaining five experiments may either be performed from the above list or as per the scope of the syllabus



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Course Code:		Course Title								L	T	P -	Credits	
BT-CSE-277A		Summer Internship -I								-	-	--	2	
										CIE		SEE		Total
									50	50		100		
Course Outcomes														
CO1	Apply fundamental programming concepts in languages (e.g. C, C++, Java, or Python) to solve real-world problems and develop functional applications.													
CO2	Implement and analyze basic data structures (e.g., arrays, linked lists, stacks, queues, trees) and algorithms (e.g., sorting, searching, recursion) in practical scenarios to identify and solve problems using algorithmic thinking and logic-based approaches.													
CO3	Understand software development life cycle (SDLC) and utilize software development tools and environments (e.g., IDEs, version control with Git/GitHub) for collaborative coding and project management.													
CO4	Write clean, modular, and well-documented code by following software engineering principles and industry coding standards.													
CO5	Demonstrate effective communication of technical concepts through documentation and presentations, while demonstrating professional ethics, time management, and collaboration during project execution.													
Course Outcomes (CO) to Programme Outcomes (PO) Mapping (Scale 1: Low, 2: Medium, 3: High)														
Cos\POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	-	-	-	-	-	-	2	2	2
CO2	3	3	2	3	2	-	-	-	-	-	-	2	2	2
CO3	2	2	2	-	2	-	-	-	2	-	-	3	3	3
CO4	2	2	-	3	2	-	-	-	1	2	2	2	2	2
CO5	-	-	-	-	1	-	-	2	3	3	2	2	2	3
Avg	2.5	2.5	2	2	2.25	-	-	2	1.5	1.25	2	2	2	2

B.Tech. student will be required to attend Inter/Intra Institute Internship-1 of 3 to 4 weeks duration after IInd semester and with the following objectives

- Provide possible opportunities to learn, understand and sharpen the real time technical / managerial skills required at the job.
- Exposure to the current technological developments relevant to the subject area of training.

The students are required to be involved in one or more of the followings Inter/ Intra Institutional activities viz; Training with higher Institutions; Soft skill training organized by Training and Placement Cell of the respective institutions; contribution at incubation/ innovation /entrepreneurship cell of the institute; participation in conferences/ workshops/ competitions etc.; Learning at Departmental Lab/ Tinkering Lab/ Institutional workshop; Working for consultancy/ research project within the institutes and Participation in all the activities of Institute's Innovation Council, eg; IPR workshop/Leadership Talks/ Idea/ Design/ Innovation/ Business Completion/ Technical Expos etc. The activity or set of activities shall, in each case, be recommended by Departmental Training Coordinator/In-charge, if any, and/or Class In-charge of the IInd year and approved by Head of the Department.

A student will write daily report which will cultivate the habit of documentation of daily observation. After completion of Internship, the student shall prepare a comprehensive report to indicate what he has observed and learnt in the training period. The student may contact Training/Internship In-charge/ Faculty Mentor to assign special topics and problems and should prepare the final report on the assigned topics. Daily diary will also help



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to a great extent in writing the Internship report since much of the information would already be incorporated by the student into the daily diary.

The Internship report will be evaluated on the basis of following criteria:

- i. Originality (A signed certificate to that effect to be attached by the student in his/her Internship report)
- ii. Adequacy and purposeful write-up.
- iii. Organization, format, drawings, sketches, style, language etc. Variety and relevance of learning experience.
- iv. Practical applications, relationships with basic theory and concepts taught in the course.

This Internship will be evaluated on the basis of a presentation made by the students by the following committee as per guidelines stipulated in the Ordinance

- i. Head of the Department
- ii. In charge Internship
- iii. Class In charge



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Syllabus

4th Semester



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B. Tech. Computer Science Engineering (Semester-IV)										
Scheme of Studies & Examination (w.e.f. session 2025-2026)										
Course No.	Course Title	Period (s)			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maximum Marks	Maximum Marks	Maximum Marks	
Engineering Science Courses										
BT- ECE-202A	Microprocessor based System Design	3	-	-	3	3	40	60	100	3 Hrs
BT- ECE-272A	Microprocessor based System Design	-	-	2	2	1	50	50	100	3 Hrs
Professional Core Courses										
BT-CSE-202A	Discrete Structures	3	1	-	4	4	40	60	100	3 Hrs
BT-CSE-204A	Computer Organization & Architecture	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-206A	Operating System	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-208A	Design & Analysis of Algorithms	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-276A	Operating Systems Lab	-	-	4	4	2	50	50	100	3 Hrs
BT-CSE-278A	Design & Analysis of Algorithms Lab	-	-	4	4	2	50	50	100	3 Hrs
Total		15	1	8	24	20	300	400	700	
Honours Basket 1 for 4 th Semester (Computer Science & Engineering)										
BT-CSE-252A	Foundations of Machine Intelligence: Concepts, Techniques, and Applications	3	-	2	5	4	40	60	100	3 Hrs
BT-CSE-254A	Foundation of Data Science	3	-	2	5	4	40	60	100	3 Hrs
Exit Course after 4th Semester										
Course No.	Course Title	Teaching Schedule			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Max	Max	Max	
BT-CSE-305A	Software Engineering/ NPTEL/' Project	3	-	-	3	3	40	60	100	3 Hrs
BT-CSE-379A	Internship-II	-	-	-	-	3	50	50	100	3 Hrs
Total (Exit Option)		3	0	0	3	6	90	110	200	
Grand Total (with Exit Option & without Honours)		36	2	24	62	52	880	1120	2000	
Note: 4 weeks Internship-II is mandatory at the end of 4th semester (before the commencement of 5th semester).										
Students Exiting the program after Securing 95.5.0 credit will be awarded UG Diploma of computer Science Engineering.										
Students opting for the Honours program are required to choose any one of Two Courses & complete a minimum of 4 credits from designated honours Course Basket 1.										



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Course Code				Course Title							L	T	P	Credits
BT-ECE-202A				Microprocessors Based System Design							3	-	-	3
											CIE	SEE	Total	
											40	60	100	
Pre-requisites: Basics of digital electronics and computer architecture														
Course Outcomes														
CO1	To understand the detailed architecture of microprocessor 8086													
CO2	To implement the interfacing of memories to 8086 Microprocessor and study of interrupt													
CO3	To classify and explain the instruction set and programming concepts in assembly language.													
CO4	To apply the programming tools to Interface peripherals with microprocessor													
Cos/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO2	3	3	3	2	-	-	-	-	-	-	-	-	3	2
CO3	3	3	3	2	-	-	-	-	-	-	-	-	3	2
CO4	3	3	3	2	-	-	-	-	-	-	-	-	3	2
Avg	3	3	3	2	-	-	-	-	-	-	-	-	3	2

Unit-1 8086 CPU ARCHITECTURE

Contact Hours: 12

Introduction to basic computer architecture, Functional units, Basic operational concepts, Harvard architecture, Von-Neumann Architecture, 8086 Block diagram; description of data registers, address registers; pointer and index registers, PSW, Queue, BIU and EU. 8086 Pin diagram descriptions. Generating 8086 CLK and reset signals using 8284. WAIT state generation. Microprocessor BUS types and buffering techniques, 8086 minimum mode and maximum mode CPU module.

Unit-2 MAIN MEMORY SYSTEM DESIGN

Contact Hours: 10

Memory devices, 8086 CPU Read/Write timing diagrams in minimum mode and maximum mode. Units of memory capacity, concept of memory banking, Address decoding techniques. Interfacing different configurations of SRAMS, ROMS/PROMS with 8086 based CPU module. Interfacing and refreshing DRAMS. Interrupts: 8086 Interrupt mechanism; interrupt types and interrupt vector table. Applications of interrupts

Unit-3 8086 INSTRUCTION SET AND PROGRAMMING TECHNIQUES

Contact Hours: 10

Instruction formats, addressing modes, Data transfer instructions, string instructions, logical instructions, arithmetic instructions, transfer of control instructions; process control instructions; Assembler directives. Writing assembly Language programs for logical processing, arithmetic processing, timing delays; loops, data conversions

Unit-4 BASIC I/O INTERFACE

Contact Hours: 12

I/O Intel's 8255 - description and interfacing with 8086. Interfacing of 0808/0809 ADCs and 0800 DACs, Generation of different waveforms. Interfacing of stepper motor, Interfacing LED displays, Interfacing Keyboards, optical encoder with 8086. Universal synchronous asynchronous receiver transmitter- 8251: describing and interfacing with 8086. Intel's 8259. DMA operation, Intel's 8237.

Text Books:

1. D.V. Hall, Microprocessors and Interfacing, McGraw Hill 2nd ed.
2. Barry B. Brey, "The Intel Microprocessor 8086/8088, 80186", Pearson Education, Eighth Edition, 2009
3. Morris Mano, "Computer System Architecture", PHI.



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Reference Books:

1. Liu, Gibson, "Microcomputer Systems: The 8086/88 Family", 2nd Edition, PHI, 2005
2. Kenneth Ayala, "The 8086 Microprocessor: Programming & Interfacing the PC", Cengage Learning, Indian Edition, 2008
3. Kip Irvine, "Assembly language for IBM PC", PHI, 2nd Edition, 1993
4. Peter Abel, "Assembly language programming", Pearson Edu, 5th Edition, 2002

NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions is to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



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Course Code				Course Title						L	T	P	Credits	
BT-ECE-272A				Microprocessors Based System Design Lab						0	0	2	1	
										CIE	SEE	Total		
										50	50	100		
Pre-requisites: Basics of digital electronics and computer architecture														
Course Outcomes														
CO1	Implement Fundamental Arithmetic and Logical Operations in Assembly Language													
CO2	Apply Memory Manipulation Techniques in Assembly Language													
CO3	Implement String and Data Processing Operations in Assembly Language													
CO4	Develop Algorithmic Solutions for Problem Solving in Assembly Language													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
Cos/P Os	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	2	2	1	-	1	3	
CO2	3	2	3	1	-	-	-	2	2	1	-	1	3	2
CO3	3	2	3	1	-	-	-	2	2	1	-	1	3	2
CO4	3	2	3	1	-	-	-	2	2	1	-	1	3	2
Avg	3	2	3	1	-	-	-	2	2	1	-	1	3	2

List of Experiments:

- Write an assembly language program to add/sub two 8-bit, 16-bit and 32-bit numbers.
- Write an assembly language program to multiply two 16 bit unsigned/signed numbers.
- Write an assembly language program to divide two unsigned/signed numbers (32/16, 16/8, 16/16, 8/8).
- Write an assembly language program to find 1's and 2's complement of a number.
- Write an assembly language program to evaluate the following equation:

$$x = \frac{(6 \times 8) + 9 - 7}{5}$$
- Write an assembly Language program to multiply 2, 8-bit numbers using repetitive addition. Assume the multiplicand is saved at location DS:0050 and the multiplier is available at location DS:0060.
- Write an assembly language program to save 55H on 10 memory locations starting from DS:200H onwards.
- Write an ALP to copy a chunk of 5 bytes of data starting from memory location DS:0100H to memory location DS:0110H
- Write an ALP to exchange two chunks of data, 8 bytes each. First data begins from memory location DS:0100H and the second data begins from memory location DS:0120H.
- Write an assembly language program to add 10 Byte of data available in the memory from location DS:0050H onwards and save the result in memory location DS:0020H.
- Write an assembly language program to convert the following two unpacked BCD numbers into a packed BCD number:
 (a) 05H
 (b) 08H
- Write an assembly language program to search the presence of a given byte in a list of 20 byte available in the memory starting from the memory location DS:0100H.
- Write an assembly language program to count the number of 1's in a given byte. Save the count in DL register.



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14. Write an assembly language program to check the parity of a given byte. If the parity is odd save 0FH in DL register. If the parity is even save 0F0H in the DL register.
15. Write an assembly language program to find smallest /largest number from array of n numbers.
16. Write an assembly language program to arrange numbers in array in ascending / descending order.
17. Write an assembly language program to convert Hex to Decimal, Decimal to Hex.
18. Write an assembly language program to compare two strings using string instructions / without using string instructions.
19. Write an assembly language program to display string in reverse order, string length, concatenation of two strings.
20. Write an assembly language program to find factorial of a number.
21. Write an assembly language program to transfer the data between 8086 Microprocessor and Peripheral Device Using 8255 Programmable Peripheral Interface.

Note: Students have to perform at least 15 experiments from the given list of practical.



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Course	Course Title										L	T	P	Credits
BT-CSE-202A	Discrete Structures										3	1	0	4
											CIE	SEE	Total	
											40	60	100	
Course Outcomes														
CO1	Apply the fundamental principles of set theory and propositional logic to solve problems using Venn diagrams, truth tables, and mathematical induction.													
CO2	Analyze various types of relations and functions, evaluate their properties, and solve combinatorial problems using generating functions, recurrence relations.													
CO3	Demonstrate understanding of graph structures and apply traversal algorithms to solve problems involving paths.													
CO4	Analyze algebraic structures such as groups, monoids, and rings, and evaluate their													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
Cos/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO 1	PSO 2
CO1	3	2	2	2	2	-	-	-	-	-	-	1	3	2
CO2	3	2	3	3	2	-	-	-	-	-	-	2	3	3
CO3	3	2	3	2	2	-	-	-	-	-	-	2	3	3
CO4	3	2	2	2	2	-	-	-	-	-	-	1	3	2
Avg	3	2	2.5	2.2	2							1.5	3	2.5

Course Outline

Unit 1: Set Theory & Logic Fundamentals

Contact Hours: 10

Sets and subsets, Venn Diagrams, Operations on sets, Laws of Set Theory, Power Sets and Products, Partition of sets, The Principle of Inclusion-Exclusion. Logic: Propositions and Logical operations, Truth tables, Equivalence, Implications, Laws of Logic, Normal forms, Predicates and quantifiers, Mathematical Induction.

Unit 2: Relations and Functions

Contact Hours: 12

Relations, Product sets, partitions and diagrams, paths in relations, properties of relations, equivalence and partially ordered relations, Transitive closure and Warshall's algorithm Functions and Combinatorics, Definitions and types of functions: injective, surjective and bijective, Composition, identity and inverse, Review of Permutation and combination-, Pigeon hole principle, Generating function-Recurrence relations.

Unit 3: Graph Theory

Contact Hours: 10

Basic terminology, types of graphs, subgraphs, representation of graphs, graph transversal algorithm: BFS and DFS, Eulerian and Hamiltonian paths, Graph colouring and planner graphs.

Unit 4: Algebraic Structures

Contact Hours: 10

Algebraic structures with one binary operation -semi groups, monoids and groups, Product and quotient o algebraic structures, Isomorphism, homomorphism, automorphism, Cyclic groups, Normal sub group, Ring homomorphism and Isomorphism.

Textbook/s:

1. C.L Liu, "Elements of Discrete Mathematics", McGraw Hill, Reprinted 2000, 1985.
2. B Kolman RC Busby, S Ross, "Discrete Mathematical Structures", PHI Pvt. Ltd.
3. Satinder Bal Gupta, "Discrete Mathematics and Structures", Laxmi Publications, Reprint 2013.



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Reference Books:

1. Tremblay J.P, and Manohar R., “Discrete Mathematical Structures with Applications to Computer Science”, by McGraw Hill Book Company, 1975, International Edition, 1987.
2. Ralph P., Grimaldi, “Discrete and Combinatorial mathematics”, Addison-Wesley Publishing Company, Reprinted in 1985.
3. Kenneth H.Rosen, “Discrete Mathematics and its Applications”, McGraw Hill Book Company, 1999. Sections: 7.1 to 7.5.

NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions is to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



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Course Code	Course Title									L	T	P	Credits	
BTCSE204A	Computer Organization & Architecture									3	-	-	3	
										CIE	SEE	Total		
										40	60	100		
Purpose	Student will be able to understand the basic concepts of computer architecture and organization, and understand the key skills of constructing cost-effective computer systems.													
Course Outcomes														
CO1	To study the functional unit of the processor such as the register file and arithmetic logical unit and with the basic of system topic													
CO2	To analysis the design trade-offs in designing and understanding the construction of a computer processor.													
CO3	To analyze the CPU design including the RISC/CISC architectures.													
CO4	To demonstrate the basic knowledge of I/O devices and interfacing of I/O devices with computer, Direct Memory Access Transfer and CPU-IOP communication.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
Cos/	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	-	-	-	-	-	1	-	-	-	-	-
CO2	3	2	2	2	1	-	-	-	1	1	-	-	2	2
CO3	3	2	2	-	1	1	-	-	1	1	1	2	-	2
CO4	3	2	2	1	1	-	-	-	1	1	1	2	2	2
Avg	3	2	2	0.75	1	1	-	-	1	1	1	2	2	2

Course outline

Unit- I: Data representation and Computer arithmetic:

Contact Hours:10

Introduction to Computer Systems, Organization and architecture, evolution and computer generations; Registers and type of registers, Buses and types of buses, Multiplication using Booth's algorithm and Division using restoring and non-restoring algorithms. Floating point representation with IEEE standards and its arithmetic operations. Memory Organization: Memory Hierarchy, Main Memory, Auxiliary Memory: magnetic disk, magnetic tape and optical disks, Cache Memory: concept and design issues & performance.

Unit-II: Basic Computer organization and Design:

Contact Hours:10

Processor organization, general registers organization, stack organization, computer instructions, instruction cycle: Fetch and Decode, Register reference instructions; Memory reference instructions. Input, output and Interrupt: configuration, instructions, Program interrupt, Interrupt cycle, Micro programmed Control Organization, Control Memory, address sequencing, Micro program Example, micro instruction format, Horizontal Vs. Vertical micro-programming, design of control Unit, micro program sequencer, Hardwired v/s Micro-programmed Control Unit.

Unit-III: Central Processing Unit:

Contact Hours:10

General register organization, stack organization, instruction formats (Zero, One, Two and Three Address Instruction), addressing modes, Data transfer and manipulation, Program control. CISC and RISC: features and comparison. Flynn's taxonomy, Pipelining, Instruction Pipeline, and Basics of vector processing and Array Processors.



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Unit-IV: Input-output organization:

Contact Hours:11

I/O interface. I/O Bus and interface modules, I/O versus Memory Bus. Asynchronous data transfer: Strobe control, Handshaking, Asynchronous serial transfer. Modes of Transfer: Programmed I/O, Interrupt driven I/O, Priority interrupt; Daisy chaining, Parallel Priority interrupt. Direct memory Access, DMA controller and transfer. Input output Processor, CPU-IOP communication, Serial communication.

Suggested Books:

1. William Stallings, “Computer Organization and Architecture – Designing for Performance”, Sixth Edition, Pearson Education, 2003.
2. Morris Mano, M., “Computer System Architecture,” 3/e, Pearson Education, 2005.
3. John P. Hayes, “Computer Architecture and Organization,” 3/e, TMH, 1998.
4. David A. Patterson and John L. Hennessy, “Computer Organization and Design: The Hardware/Software interface”, Third Edition, Elsevier, 2005.
5. V.P. Heuring, H.F. Jordan, “Computer Systems Design and Architecture”, Second Edition, Pearson Education, 2004.
6. Carl Hamacher, Zvonko Vranesic and Safwat Zaky, “Computer Organization”, Fifth Edition, Tata McGraw Hill, 2002

NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions is to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



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Course Code	Course Title	L	T	P	Credits
BT-CSE-206 A	Operating System	3	0	0	3
		CIE	SEE		Total
		40	60		100

Course Outcomes

CO1	To understand the structure and functions of Operating system.
CO2	To learn about process management, scheduling algorithms and Process Synchronization.
CO3	To understand the concept of deadlock and memory management.
CO4	The students will be able to understand the concept of file system and disk management.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

Cos/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PS O2
CO1	3	3	3	2	-	-	-	-	-	1	-	2	2	2
CO2	3	3	3	2	1	-	-	-	-	1	-	2	3	1
CO3	3	3	2	1	1	-	-	-	-	1	-	1	2	-
CO4	3	3	3	2	1	-	-	-	-	2	-	2	1	2
Avg	3	3	2.75	1.75	1	-	-	-	-	1.25	-	1.75	2	1.25

Course Outline

Unit 1: Introduction to operating System

Contact Hours:10

Introduction: Introduction to OS. Operating system functions, Different types of O.S.: batch process, multiprogrammed, time-sharing, real-time, distributed, parallel. System Structure: Computer system operation, I/O structure, storage structure, storage hierarchy, different types of protections, operating system structure (simple, layered, virtual machine), O/S services, system calls.

Unit 2: Process Management and Synchronization

Contact Hours:10

CPU scheduling: scheduling criteria, preemptive and non-preemptive scheduling, scheduling algorithms, algorithm evaluation, multi-processor scheduling. Threads: overview, benefits of threads, user and kernel threads. Process Management: Concept of processes, process states, process control, co-operating processes, inter-process communication. Process Synchronization: background, critical section problem, critical region, synchronization hardware, Classical problems of synchronization, semaphores.

Unit 3: Deadlock and Memory management

Contact Hours:10

Deadlocks: Concept of deadlock, deadlock characterization, deadlock prevention, deadlock avoidance, deadlock detection, recovery from deadlock. Memory Management: background, logical vs. physical address space, contiguous memory allocation, paging, segmentation, segmentation with paging. Concept of fragmentation. Virtual Memory: background, demand paging, concept of page replacement, page replacement algorithms, allocation of frames, thrashing.

Unit 4: File system and Disk management

Contact Hours:12

File Systems: file concept, file organization and access methods, allocation methods, directory structure, free space management I/O Management: I/O hardware, polling, interrupts, DMA, kernel I/O subsystem (scheduling, buffering, caching, spooling and device reservation) Disk Management: disk structure, disk scheduling (FCFS,



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SSTF, SCAN, C-SCAN) , disk reliability, disk Performance parameters Protection and Security: Goals of protection and security, security attacks, authentication, program threats, system threats, threat monitoring. Case studies: UNIX file system, Windows file system.

Text Book

1. Operating System Concepts”, Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne, Wiley
2. Operating Systems Design and Implementation” ,(Prentice Hall Software Series) Andrew S Tanenbaum and Albert S Woodhull.

Reference Books

1. Operating systems: a concept-based approach”, Dhananjay M. Dhamdhare, McGraw Hill
2. Operating Systems: Internals and Design Principles, William Stallings, Pearson

NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions is to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



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Course Code	Course Title	L	T	P	Credits
BT-CSE-208A	Design & Analysis of Algorithms	3	0	0	3
		CIE	SEE		Total
		40	60		100

Course Outcomes

CO1	To compute algorithm efficiency through asymptotic notations and solve recurrence-based problems.
CO2	To solve computational problems using dynamic programming, greedy methods, and backtracking techniques.
CO3	To analyze advanced data structures and graph algorithms for solving complex computational problems efficiently.
CO4	To analyze the efficiency of flow algorithms, sorting networks, and concepts related to computational complexity.

Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)

Cos/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	-	-	-	1	-	-	3	3	3
CO2	3	3	3	2	2	-	-	-	1	-	-	3	3	3
CO3	3	3	3	2	2	-	-	-	1	-	-	3	3	3
CO4	3	3	3	2	2	-	-	-	1	-	-	3	3	3
Avg	3	3	3	2	2	-	-	-	1	-	-	3	3	3

Course Outline

Unit 1: Algorithm Complexity and Recurrence

Contact Hours:10

Review: - Elementary Data Structures, Algorithms and their complexity (Time and Space), Analysing Algorithms, Asymptotic Notations: Big O, Omega (Ω), and Theta (Θ), Quick Sort algorithm, Merge Sort algorithm, Recurrence relations: - Methods for solving recurrence (Substitution, Recursion tree, Master theorem), Strassen multiplication

Unit 2: Algorithm Design Strategies

Contact Hours:11

Dynamic programming: - Elements, Matrix-chain multiplication, longest common subsequence, Greedy algorithms: - Introduction, Activity- Selection problem, Knapsack Problem, Huffman codes, Travelling Salesman Problem. Backtracking: Introduction, Subset-Sum Problem, N-Queen Problem

Unit 3: Advanced Data Structure and Graph Algorithms

Contact Hours:11

Advanced Data Structures: - Binary Search Tree (insertion and deletion), Splay Tree, Heap Sort algorithm, Binomial heaps, Fibonacci heaps, Review of graph algorithms: -Traversal Methods (Depth first and Breadth first search), Topological sort, strongly connected components, Minimum spanning trees- Kruskal and Prims, Single source shortest paths, Relaxation, Dijkstra's Algorithm, Bellman- Ford algorithm, Single source shortest paths for directed acyclic graphs, All pairs shortest paths- shortest paths and matrix multiplication, Floyd-Warshall algorithm.

Unit 4: Flow Algorithms and Computational Complexity

Contact Hours: 10

Flow and Sorting Networks: Flow networks, Ford-Fulkerson method, Maximum Bipartite matching, Sorting Networks, Comparison network, the zero-one principle, Bitonic sorting network, Merging networks, Computational Complexity: Basic Concepts, Polynomial vs Non-Polynomial Complexity, NP- hard & NP-complete classes.



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Suggested Books:

1. Corman, Leiserson, and Rivest: Introduction to Algorithms, 2/e, PHI
2. Horowitz, Ellis, and Sahni, Sartaj: Fundamentals of Computer Algorithms, Galgotia Publications
3. Aho, Hopcroft, and Ullman: The Design and Analysis of Computer Algorithms. Addison Wesley.
4. R. B. Patel: Expert Data Structures with C, Khanna Publications, Delhi, India, 2nd Edition 2004, ISBN 81-87325-07-0.
5. R. B. Patel and M.M.S Rauthan: Expert Data Structures with C++, Khana Publications, Delhi, India, 2nd Edition 2004, ISBN 87522-03-8

NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions is to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



Panipat Institute of Engineering and Technology
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Course Code				Course Title							L	T	P	Credits
BT-CSE-276A				Operating System Lab							-	-	4	2
											CIE	SEE	Total	
											50	50	100	
Purpose				To familiarize the students with the basics of Operating Systems.										
Course Outcomes														
CO1	To evaluate various CPU scheduling algorithms.													
CO2	To implement various classical problems of process synchronization.													
CO3	To implement various Deadlock detection methods, memory management schemes and file operation.													
CO4	To evaluate various page replacement and disk scheduling algorithm													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
Cos/	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1POs	3	3	3	2	2	-	-	-	-	-	-	2	2	2
CO2	3	2	2	1	2	-	-	-	-	-	-	2	2	1
CO3	3	2	2	1	2	-	-	-	-	-	-	2	2	1
CO4	3	3	3	2	2	-	-	-	-	-	-	2	2	2
Avg	3	2.5	2.5	1.5	2	-	-	-	-	-	-	2	2	1.5

List of Practical

1. Write a program to simulate the CPU scheduling algorithms: (i) FCFS (ii) SJF and (iii) SRTF
2. Write a program to simulate the CPU scheduling algorithms: (i) Preemptive Priority (ii) Non-Preemptive Priority and (iii) Round Robin
3. Write a program to implementation of process and thread (Life cycle of process): (i) Process creation and termination; (ii) Thread creation and termination
4. Write a program to implement classical problems of process synchronization using semaphores: (i) Producer consumer (ii) Reader writer and (iii) Dining philosophers'
5. Write a program to implement Banker's algorithm for dead lock avoidance.
6. Write a program to implement paging and segmentation schemes of memory management.
7. Write a program to simulate memory allocation methods: (i) Best Fit (ii) Worst Fit and (iii) FirstFit
8. Write a program to simulate the page replacement algorithms: (i) FIFO (ii) LRU and (iii) Optimal
9. Write a program to simulate the disk scheduling algorithms: (i) FCFS (ii) SSTF and (iii) SCAN
10. Write a program to simulate the disk scheduling algorithms: (i) C-SCAN (ii) LOOK (iii) C-LOOK
11. Write a program to implement file allocation strategies.
12. Write a program to simulate all file organization techniques.

NOTE: A student must to perform at least ten experiments. Seven experiments should be performed from the above list. The remaining three experiments may either be performed from the above list or as per the scope of the syllabus



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Course Code				Course Title							L	T	P	Credits
BT-CSE-278A				Design Analysis of Algorithms Lab							-	-	4	2
											CIE	SEE	Total	
											50	50	100	
Purpose				To Learn the Design Analysis of Algorithms										
Course Outcomes														
CO1	To apply suitable sorting techniques for organizing data in various computational problems.													
CO2	To solve real-world optimization and decision problems using dynamic programming and backtracking strategies.													
CO3	To design and analyze algorithms for solving graph-based traversal, ordering, and optimization problems.													
CO4	To evaluate shortest path algorithms for determining optimal solutions in weighted graph problems.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	1	-	-	1	-	-	3	3	3
CO2	3	3	3	2	2	-	-	-	1	-	-	3	3	3
CO3	3	3	3	2	2	-	-	-	1	-	-	3	3	3
CO4	3	3	3	2	2	-	-	-	1	-	-	3	3	3
Avg	3	3		2	2	-	-	-	1	-	-	3	3	3

List of Experiment

1. Implement the Quick Sort algorithm to sort a given list of elements.
2. Develop a program to implement Merge Sort for sorting a list of elements.
3. Implement a solution for the 0/1 Knapsack problem to determine the maximum profit achievable within the given weight constraint.
4. Develop a program that calculates the most efficient way to multiply a chain of matrices by determining the optimal parenthesizing using dynamic programming.
5. Develop a Dynamic Programming solution to find and display the Longest Common Subsequence of two given strings.
6. Implement the Traveling Salesman Problem to find the minimum-cost tour visiting all cities once.
7. Find a subset of a given set $S = \{s_1, s_2, \dots, s_n\}$ of n positive integers whose sum is equal to a given positive integer d . For example, if $S = \{1, 2, 5, 6, 8\}$ and $d = 9$ there are two solutions $\{1, 2, 6\}$ and $\{1, 8\}$. A suitable message is to be displayed if the given problem instance doesn't have a solution.
8. Develop a program to place N queens on a chessboard using backtracking.
9. Implement Graph Coloring using Backtracking
10. Create a program that uses BFS to systematically traverse all vertices of a graph starting from a given node.
11. Write a program to explore all nodes of a graph using Depth-First Search (DFS) starting from a given node.
12. Develop a program to find a topological order of vertices in a directed acyclic graph using DFS.
13. Implement Kruskal's algorithm to determine the Minimum Cost Spanning Tree of an undirected graph.
14. Find the Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.



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15. From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.
16. Write a program to find the shortest paths between all pairs of vertices in a weighted graph using the Floyd-Warshall algorithm.

NOTE: A student must to perform at least ten experiments. Seven experiments should be performed from the above list. The remaining three experiments may either be performed from the above list or designed & set by the concerned institution as per the scope of the syllabus.



Honours Basket 1: 4th Semester

B. Tech. Computer Science Engineering (Semester-IV) w.e.f. 2025-26										
Course No.	Course Title	Period (s)			Hours/ Week	Credit (s)	Continuous Internal Evaluation (CIE)	Semester End Examination (SEE)	Total Marks (CIE+SEE)	Duration of Exam (Hours)
		L	T	P			Maximum Marks	Maximum Marks	Maximum Marks	
Honours Basket 1 for 4 th Semester (Computer Science & Engineering)										
BT-CSE-252A	Foundations of Machine Intelligence: Concepts, Techniques, and Applications	3	-	2	5	4	40	60	100	3 Hrs
BT-CSE-254A	Foundation of Data Science	3	-	2	5	4	40	60	100	3 Hrs
Students opting for the Honours program are required to choose any one of Two Courses & complete a minimum of 4 credits from designated honours Course Basket 1.										



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Course Code				Course Title						L	T	P	Credits		
BT-CSE-252A				Foundations of Machine Intelligence: Concepts, Techniques and Applications						3	0	0	-		
										CIE		SEE		Total	
										40		60		100	
Course Outcomes															
CO1		Describe the evolution, scope, and applications of Artificial Intelligence and the role of intelligent agents in dynamic environments.													
CO2		Apply logical reasoning techniques and knowledge representation methods using propositional and predicate logic for problem-solving in AI systems.													
CO3		Analyze different types of data and perform data analysis using statistical and exploratory techniques to extract meaningful insights.													
CO4		Evaluate machine learning approaches and emerging AI technologies like XAI, Edge AI, and AI for Social Good through real-world case studies.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
Cos/P Os	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PS O2	
CO1	3	2	-	-			2		2	2		2	2	2	
CO2	3	3	2	-	2				2	2		2	3	3	
CO3	3	3	-	2	2				2	2		2	2	2	
CO4	2	3	3	3	3	1	2	2	2	2		2	3	3	

Course Outline:

Unit: 1 Introduction to AI and Intelligent Agents

Contact Hours:10

AI Essentials: Definition and scope of AI, History and evolution of AI, AI vs. Human Intelligence, AI in engineering and technology. Intelligent Agents and Environment Interaction: Definition and Types of Intelligent Agents, Agents and Environments: How agents perceive and act. Knowledge

Unit: 2 Knowledge Representation and Data Analysis

Contact Hours:12

Representation and Reasoning: Introduction to Knowledge Representation, Logic-based approaches: Propositional and Predicate Logic, Rule-based Systems and Expert Systems, Introduction to Data and its analysis: Types of data: Structured, Semi-structured, Unstructured, Data Streams, Statistical Data types, Sampling. Data Analysis Methods: Descriptive Analysis, Exploratory Analysis, Inferential Analysis, Predictive Analysis.

Unit: 3 Machine Learning Essentials

Contact Hours:12

Introduction to Machine Learning, Types of Learning (Supervised, Unsupervised, Semi-Supervised, and Reinforcement Learning), Introduction to Neural Networks and Deep Learning. Recent Advancements in AI: Explainable AI (XAI), Edge AI, Generative AI, AI for Social Good
 (Applications of AI in healthcare, education, climate change, and sustainability), AI Security
 (Protecting AI systems from attacks and ensuring secure AI development)

Unit: 4 Case studies of AI-driven innovations

Contact Hours:10

Smart Healthcare, Smart Agriculture, Smart Speakers, Self-Driven cars, Smart Education

Text Book

1. Artificial Intelligence: A Modern Approach - Stuart Russell & Peter Norvig, Pearson Publishers 4th Edition
2. Machine Learning - Tom M. Mitchell, McGraw-Hill Publishers
3. Pattern Recognition and Machine Learning-Christopher M. Bishop, Springer Publishers



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Reference Books

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow- Aurélien Géron, O'Reilly Media Publisher
2. Understanding Machine Learning: From Theory to Algorithms- Shai Shalev-Shwartz, Shai Ben-David, Cambridge University Press Publishers

Online Learning Resources/URLs:

1. Learn Intro to Machine Learning | Kaggle
2. <https://www.coursera.org/learn/machine-learning>
3. Machine Learning| Google for Developers

NOTE:

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2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



Panipat Institute of Engineering and Technology
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Course Code					Course Title							L	T	P	Credits
BT-CSE-272A					Foundations of Machine Intelligence Lab							-	-	4	2
												CIE	SEE	Total	
												40	60	100	
Course Outcomes															
CO1	Apply Python libraries (NumPy, Pandas, Matplotlib, Seaborn) for data handling and visualization														
CO2	Perform data cleaning, preprocessing, and summary statistics for EDA														
CO3	Visualize data to understand distributions, relationships, and patterns														
CO4	Utilize Python libraries for data wrangling tasks.														
CO5	Implement basic predictive modelling using linear regression														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
Cos/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
CO1	3	2	-	-	3	-	-	-	-	-	-	2	3	3	
CO2	3	3	-	2	3	-	-	-	-	-	-	2	2	2	
CO3	2	3	-	-	3	-	-	-	-	-	-	2	3	3	
CO4	3	2	-	-	3	-	-	-	-	-	-	2	2	2	
CO5	3	3	2	2	3	-	-	-	-	-	-	2	3	3	
Avg	3	3	2	2	3	-	-	-	-	-	-	2	2	2	

List of Experiment:

1. Implementation of Python Basic Libraries such as Statistics, Math, NumPy and SciPy.
2. Implementation of Python Libraries for ML application such as Pandas and Matplotlib.
3. Write a python program to perform Basic operations, arrays, data frames, and data cleaning.
4. Write a python program to view structure, dimensions, and summary statistics using Panda's libraries.
5. Write a python program to creation and loading different types of datasets using the required libraries.
6. Write a python program to visualize and detect anomalies in the dataset.
7. Write a python program to compute mean, median, mode, variance, standard deviation using datasets.
8. Write a python program to perform handling nulls, encoding categorical data, scaling, train-test split.
9. Write a python program to demonstrate various data pre-processing techniques for a given dataset.
10. Write a python program to explore feature distributions using histograms, boxplots, and count plots method.
11. Write a python program to explore relationships between variables using scatter plots and heatmaps.
12. Write a Python program to implement simple linear regression and plot the graph.
13. Write a python program to predict continuous values, residual analysis, R^2 score of linear Regression.
14. Write a python program to perform to predict values (e.g., house prices or marks) using linear regression.
15. Write a Python program to implement Simple Linear Regression for iris using sklearn and plot the confusion matrix.



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Reference Books:

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow by Aurélien Géron, Publisher: O'Reilly Media.
2. Python Machine Learning by Sebastian Raschka, Vahid Mirjalili, Publisher: Packt Publishing
3. Machine Learning Using Python. Manaranjan Pradhan, U Dinesh Kumar, Publisher: Wiley India

NOTE: A student has to perform at least ten experiments. Seven experiments should be performed from the above list. Remaining three experiments may either be performed from the above list or as per the scope of the syllabus



Panipat Institute of Engineering and Technology
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Course Code				Course Title						L	T	P	Credits		
BT-CSE-254A				Foundation of Data Science						3	0	0	-		
										CIE		SEE		Total	
										40		60		100	
Course Outcomes															
CO1	Understand the fundamentals and processes of data science.														
CO2	Learn to describe data effectively within the data science workflow.														
CO3	Explore relationships between different data variables.														
CO4	Utilize Python libraries for data wrangling tasks.														
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)															
COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
CO1	3	2	-			-	-	-	-	-	-	2	-	-	
CO2	3	2	-		2	-	-	-	-	-	-	2	-	-	
CO3	3	3	-	2	2	-	-	-	-	-	-	2	-	-	
CO4	3	2	-		3	-	-	-	-	-	-	2	3	3	
Avg	3	2.5	-	2	1.75	-	-	-	-	-	-	2	3	3	

Course Outcomes

Unit – 1: Introduction to Data Science

Contact Hours:12

Benefits and uses of Data Science, Facets of data, Overview of the Data Science Process, Defining research goals
 Data retrieval and preparation, Exploratory Data Analysis (EDA), Model building and application development
 Data Mining and Data Warehousing.

Unit – 2: Describing Data

Contact Hours:10

Types of data and variables, Descriptive statistics: tables, graphs, averages, Measures of variability,
 Normal distributions and z-scores-measure of dispersion, Data distribution and shape, Frequency Distributions
 and Tabulation

Unit – 3: Relationships Descriptions

Contact Hours:12

Correlation and scatter plots, Computational formula for correlation coefficient, Regression analysis: regression
 lines, least squares method, Standard error of estimate, Interpretation of R^2 (coefficient of determination),
 Multiple regression equations, Regression towards the mean

Unit – 4: Python Libraries for Data Wrangling

Contact Hours:08

Basics of NumPy arrays, Aggregations and computations on arrays, Comparisons, masks, and boolean logic, Fancy
 indexing and structured arrays, Data manipulation with Pandas: indexing, selection, operations, Handling missing
 data Hierarchical indexing, combining datasets, Aggregation, grouping, and pivot tables.

Reference Books:

1. Introduction to Data Science: A Python Approach to Concepts, Techniques and Applications, Author: Laura Igual, Santi Seguí, Publisher: Springer Cham.
2. Python for Data Analysis, 2nd Edition by Wes McKinney Released October 2017 Publisher(s): O'Reilly Media, Inc.
3. Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter, Third Edition by Wes McKinney
4. Practical Statistics for Data Scientists” by Peter Bruce, Andrew Bruce, and Peter Gedeck



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NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions are to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



Panipat Institute of Engineering and Technology
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Course Code				Course Title								L	T	P	Credits
BT-CSE-274A				Foundation of Data Science Lab								-	-	4	2
												CI	SEE	Total	
												40	60	100	
Course Outcomes															
CO1		Apply Python programming and libraries such as NumPy and Pandas to manipulate and analyze data.													
CO2		Perform data retrieval, cleaning, and exploratory data analysis (EDA) on real-world													
CO3		Apply descriptive statistics and correlation techniques for data interpretation and visualization													
CO4		Develop and evaluate regression models for making predictions from structured data.													
Course Outcomes (CO) to Programme Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High															
Cos/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
CO1	3	2	-		3	-	-	-	-	-	-	2	2	2	
CO2	3	3	-	2	3	-	-	-	-	-	-	2	2	2	
CO3	3	3	-	2	3	-	-	-	-	-	-	2	2	2	
CO4	3	3	-	2	3	-	-	-	-	-	-	2	3	3	
Avg	3	3	2	2	3	-	-	-	-	-	-	2	3	3	

List of Practicals:

1. Setup Python environment and write simple Python scripts for data input and basic operations.
2. Retrieve datasets and perform initial cleaning (remove duplicates, handle null values, etc.).
3. Explore the data using graphs (histograms, boxplots, scatterplots) and descriptive statistics.
4. Use NumPy for numerical computation, indexing, aggregations, and boolean logic.
5. Perform slicing, filtering, and column operations on real datasets using Data Frames & Series.
6. Handle missing data, perform hierarchical indexing, and combine multiple datasets using Pandas.
7. Analyze the distribution and shape of data using mean, median, mode, standard deviation, and z-scores.
8. Identify relationships between variables using correlation and fit simple regression models.
9. Build multivariate regression models and interpret R^2 for predictive accuracy.
10. Summarize and reshape data using group-by and pivot_table functions in Pandas.

Text Books/ Reference Books

1. Python for Data Analysis" by Wes McKinney, Publisher: O'Reilly Media, 2nd Edition
2. Python Data Science Handbook" by Jake VanderPlas, Publisher: O'Reilly Media, 1st Edition
3. Hands-On Data Analysis with Pandas" by Stefanie Molin, Publisher: Packt Publishing, 2nd Edition

NOTE: A student has to perform at least ten experiments. Seven experiments should be performed from the above list. Remaining three experiments may either be performed from the above list or as per the scope of the syllabus



Minor Degree in Web Development

Department of Computer Science & Engineering offering Minor Degree in Web Development to other departments (w.e.f. 2025-26)										
Semester	Course No.	Course Title	Period(s)			Credit (s)	CIE	SEE	Total	Duration
			L	T	P		Max	Max		
4th Sem	BT-CSE-212A	Foundations of Web Design and Development	3			3	40	60	100	3 Hrs
	BT-CSE-272A	Foundations of Web Design and Development Lab			2	1	50	50	100	3 Hrs
5th Sem	BT-CSE-311A	Database Technologies for Web Applications	3			3	40	60	100	3 Hrs
	BT-CSE-371A	Database Technologies for Web Applications Lab			2	1	50	50	100	3 Hrs
6th Sem	BT-CSE-306A	Dynamic web Application using PHP	3			3	40	60	100	3 Hrs
	BT-CSE-380A	Dynamic Web Application using PHP lab			2	1	50	50	100	3 Hrs
7th Sem	BT-CSE-403A	Advanced Web Frameworks	3			3	40	60	100	3 Hrs
	BT-CSE-481A	Minor project-1			4	2	50	50	100	3 Hrs
8th Sem	BT-CSE-478A	Project-II			6	3	50	50	100	3 Hrs
Total			12		16	20	410	490	900	



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Course Code	Course Title	L	T	P	Credits									
BT-CSE-212A	Fundamentals of Web Designing and Development	3	0	2	3									
			CIE	SEE	Total									
			40	60	100									
Course Objective	To provide students with a strong foundation in web design and development, covering the basics of HTML, CSS, and JavaScript. Students will learn how to create static and interactive web pages, and understand the key concepts needed for building modern web applications													
Course Outcomes (CO)														
CO1	Understand the fundamentals of web Designing and development													
CO2	Develop structured and semantically meaningful web pages.													
CO3	Apply CSS techniques to style web pages, utilizing layout models													
CO4	Utilize JavaScript to add interactivity to web pages,													
CO5	Implement advanced JavaScript features, perform form validation, handle asynchronous data fetching.													
Course Outcomes (CO) to Program Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High														
Cos/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2					2		2		2		
CO2	3		3		2					3		2		
CO3	3		3							3		2		
CO4	2		3	3	3		2	2				3		
CO5		3		2	2		3	3				2		
AVG	2		2		2							2		

Course Outline

Unit I: Fundamentals of Web Designing and Development Contact Hours: 10

Overview of Web Designing and Development, Client-Server Architecture and Web Browsers; Introduction to Web Standards and Accessibility, Understanding the Structure of Web Pages, Web Development Tools and Environments, Text Editors (VS Code, Sublime Text, etc.), Web Browser Developer Tools

Unit II: Basics of HTML Tags Contact Hours: 10

HTML Basics and Advanced Techniques, Introduction to HTML Structure (Tags, Attributes, Elements), Text Formatting, Links, and Lists, Images, Audio, and Video Embedding, Tables, Forms, and Input Elements, Semantic HTML: <header>, <footer>, <article>, etc., accessibility Considerations in HTML

Unit III: User Experience (UX) and User Interface (UI) Principles CSS for Layout and Design

Contact Hours: 12

Introduction to CSS (Selectors, Properties, Values), Box Model (Padding, Margin, Border, Content), Styling Text, Colors, and Backgrounds, CSS Positioning (Static, Relative, Absolute, Fixed), Flexbox and Grid Layout Systems, Responsive Web Design (Media Queries and Mobile-First Design, CSS Transitions and Animations Maintaining Consistent Design (CSS Frameworks like Bootstrap)

Unit IV: Introduction to JavaScript for Interactivity and SEO Basics

Contact Hours: 10

JavaScript Basics (Variables, Data Types, Operators, Loops) Functions, Arrays, and Objects, Validating Forms with JavaScript, DOM Manipulation: Accessing and Modifying HTML Elements, Event Handling (Click, Hover, etc.)



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Introduction to JavaScript Libraries (e.g., jQuery), SEO Basics for Web Pages

Reference Books:

1. Web Development and Design Foundations with HTML5 by Terry Felke-Morris, publisher Pearson
- HTML and CSS QuickStart Guide by David Durocher publisher Clyde Bank Media LLC

Recommended Tools & Resources:

1. Text Editors: Visual Studio Code, Sublime Text
2. Browsers: Google Chrome, Firefox (with Developer Tools)

NOTE:

1. For the semester examination, nine questions are to be set by the examiner. Question no. 1, containing 5-7 short answer type questions, will be compulsory & based on the entire syllabus. Rest of the eight questions is to be set by setting two questions from each of the four units of the syllabus. The candidates will be required to attempt five questions in all, selecting one from each unit AND Question no. 1. All questions will carry equal marks.
2. The students will be allowed to use non-programmable scientific calculator. However, sharing / ex-change of calculator or any other items are prohibited in the examinations. No programmable calculators, mobile phones or other electrical/ electronic items are allowed in the examination.



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Course Code	Course Title	L	T	P	Credits									
BT-CSE-272A	Fundamentals of Web Designing and Development Lab		0	4	2									
			CIE	SEE	Total									
			50	50	100									
Course Objective	To provide students with a strong foundation in web design and development, covering the basics of HTML, CSS, and JavaScript. Students will learn how to create static and interactive web pages, and understand the key concepts needed for building modern web applications													
Course Outcomes (CO)														
CO1	Design structured and accessible web pages using HTML5 semantic elements and media components.													
CO2	Style and layout web pages effectively using CSS, Flexbox, Grid, and implement responsive design using media queries													
CO3	Build and validate interactive web forms using HTML and JavaScript for client-side input validation.													
CO4	Integrate HTML, CSS, and JavaScript components to develop a functional, responsive Student Event Management System.													
Course Outcomes (CO) to Program Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2			3							2	2	2
CO2	2	2	2		3							2	2	2
CO3	2	2	2		3							2	2	2
CO4	2	3	3	2	3				2	2	2	2	2	2
AVG	2	2.25	1.75	2	2.75				2	2	2	2	2	2

Lab Experiments

Web Application Goal: "Student Event Management System"

1. Introduction to Basic HTML Structure

Statement: Set up a basic folder structure (/css, /js, /images) and create the index.html file for the Student Event Management System. Implement the standard HTML5 skeleton with semantic elements like <header>, <main>, <nav>, <section>, and <footer>. Add a meaningful <title> and set up navigation placeholders.

2. Designing the Navigation Bar and Home Page

Statement: Create a horizontal navigation bar using semantic elements (<nav>, , , <a>). Design the home page using proper headings, text formatting tags, and a welcome message. Add internal links to "Events," "Register," and "Contact" sections.

3. Building the Events List with HTML Lists and Formatting

Statement: Create an "Events" section that categorizes upcoming events using ordered and unordered lists. Format each event's name, date, and description using appropriate headings and paragraph tags.

4. Embedding Media Content into the Events Page

Statement: Embed a promotional video, an audio invitation, and event images in the "Events" section using <video>, <audio>, and tags with proper attributes and controls. Ensure media accessibility with alt attributes and captions.



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5. Displaying Events in a Table Format

Statement: Create a structured HTML table to display events with fields: Event ID, Name, Date, Venue, and Type. Style the table minimally using inline or internal CSS for borders and spacing.

6. Creating a Registration Form

Statement: Build a “Register for Event” form with fields for Name, Email, Phone, Event Selection (Dropdown), and Comments. Use various input types, <label>, and accessibility attributes such as aria-label, required, and <fieldset>.

7. Styling the Web Pages Using CSS

Statement: Apply internal and external CSS to style the layout, headings, paragraphs, form inputs, buttons, and links. Use the box model for spacing and add background images or colors where appropriate.

8. Layout Design Using Flexbox and Grid

Statement: Redesign the layout of the Events and Home page using Flexbox and CSS Grid. Align cards side-by-side and create responsive column layouts for better structure and visual hierarchy.

9. Making the Site Responsive with Media Queries

Statement: Enhance your site’s responsiveness by writing media queries. Adjust font sizes, element visibility, navigation layout, and image scaling for mobile devices (max width: 768px). Use a mobile-first CSS approach.

10. Adding JavaScript for Form Validation

Statement: Add client-side validation to the registration form using JavaScript. Validate name, email, and phone number fields. Show dynamic error messages using JavaScript DOM methods and prevent form submission on invalid input.

Final Integration Project

Title: Create a Student Event Management Web Application

Objective: Combine all the above programs into a single integrated PHP-based portal. Once all 12 programs are completed, students integrate them to build a responsive and accessible Student Event Management Web Application.



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Course Code				Course Title							L	T	P	Credits
BT-CSE-311A				Database Technologies for Web Applications							3	-	-	3
											CIE	SEE	Total	
											40	60	100	
Course Outcomes														
CO1	To comprehend the role of database management systems in web applications development													
CO2	To develop a database-based web applications project depending on the business requirements, considering various design issues													
CO3	To apply NOSQL database in web in web applications development													
CO4	To apply MongoDB database in web in web applications development													
CO5	To apply advanced database techniques in deployment of project													
Course Outcomes (CO) to Program Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
COs\ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	1	-	-	-	-	-	-	-	2	1	-
CO2	3	3	3	3	2	-	-	-	-	-	-	3	2	1
CO3	3	3	3	2	2	-	-	-	-	-	-	3	2	1
CO4	3	3	3	2	2	-	-	-	-	-	-	3	2	1
CO5	3	3	3	2	2	-	-	-	-	-	-	3	2	1

Course Outline

Unit 1: Fundamentals of Database Management Systems

Contact Hours: 12

Introduction to Database Concepts, Need for database systems, Characteristics of Database Approach, Classification of database management systems, Data Models, Schemas and Instances, Centralized and Client/Server Architectures for DBMSs, Database Administrator, Web Server, Basics of SQL Database, Connecting MySQL with Web Applications

Unit 2: NOSQL Database Management

Contact Hours:10

Introduction to NOSQL Database, Need of NoSQL, CAP Theorem, different NoSQL databases: Key-value data stores, Columnar families, Document databases, Graph databases

Unit 3: Mongo DB Database

Contact Hours:10

Introduction to MongoDB for Scalable Web Apps, Integrating MongoDB with Web Applications, CRUD Operations in MongoDB

Unit 4: Advanced Database Techniques and Deployment

Contact Hours:10

Database Optimization Techniques, Backup and Recovery Strategies, Data Migration and Version Control, Deploying Databases on Cloud Platforms (AWS, Heroku, MongoDB Atlas), Introduction to REST APIs and Database Communication

Textbook:

- Gerardus Blokdyk, NoSQL Databases A Complete Guide, SSTARCOoks, 2021
- Raghu Ramakrishnan, Database Management Systems, Mcgraw-Hill, 4th Edition, 2018



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Reference Books

1. R. Elmasri & S. B. Navathe, Fundamentals of Database Systems, Addison Wesley, yth Edition, 2016.
2. A. Silberschatz, H. F. Korth & S. Sudarshan, Database System Concepts, McGraw Hill, a. 7th Edition 2019.
3. Harvey M Deitel, Paul J Deitel and Tem R Nieto, Internet and World Wide Web How to Program, Pearson, 6th Edition, 2020.

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Course Code		Course Title									L	T	P	Credits
BT-CSE-371A		Database Technologies for Web Applications Lab										-	4	2
											CIE	SEE	Total	
											50	50	100	
Course Outcomes														
CO1	To understand basic DDL commands.													
CO2	To learn about DML and DCL commands													
CO3	To study and implement commands and operations in MongoDB													
CO4	To design a web application using SQL.													
CO5	To design a web application using MongoDB.													
Course Outcomes (CO) to Program Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
COs/ POS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1		2	-	-	-	-	-	-	2	2	1
CO2	3	2	1		2	-	-	-	-	-	-	2	2	1
CO3	3	2	2		2	-	-	-	-	-	-	2	2	2
CO4	3	2	3		3	-	-	-	-	-	-	3	3	2
CO5	3	2	3		3	-	-	-	-	-	-	3	2	2
Avg	3	2	2.0	-	2.4	-	-	-	-	-	-	2.8	2.2	1.6

List of Experiment:

1. Write the data definition language (DDL) SQL queries.
2. Write the data manipulation language (DML) and DCL SQL queries.
3. Write SQL queries for extracting data from more than one table.
4. Illustration of Where Clause, AND, OR operations in MongoDB.
5. Execute the Commands of MongoDB and operations in MongoDB: Insert, Query, Update, Delete and Projection. (Note: use any collection)
6. Develop a MongoDB query to select certain fields and ignore some fields of the documents from any collection.
7. Develop a MongoDB query to display the first 5 documents from the results obtained in experiment 6. [use of limit and find]
8. Execute query selectors (comparison selectors, logical selectors) and list out the results on any collection.
9. Execute query selectors (Geospatial selectors, Bit wise selectors) and list the results on any collection.
10. Create and demonstrate how projection operators (\$, \$elemmatch and \$slice) would be used in the MongoDB.
11. Execute Aggregation operations (\$avg, \$min, \$max, \$push, \$addToSet etc.) students encourage to execute several queries to demonstrate various aggregation operators)
12. Find all listings with listing_url, name, address, host_picture_url in the listings And Reviews collection that have a host with a picture url
13. Using E-commerce collection writes a query to display reviews summary.
14. Setting up MySQL and MongoDB locally and on the cloud.
15. Develop a Real-Time Chat Application using MongoDB.



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16. Deploy a Blog Application with both MySQL and MongoDB databases.

Text Books:

1. R. Elmasri & S. B. Navathe, Fundamentals of Database Systems, Addison Wesley, 7th Edition, 2016.
2. A. Silberschatz, H. F. Korth & S. Sudarshan, Database System Concepts, McGraw Hill, 7th Edition 2019.
3. Kristina Chodorow, “MongoDB: The Definitive Guide”, 2nd ed O'REILLY, 2013. 4. Manu Sharma “MongoDB Complete Guide” 1st ed, bpb publication, 2023.



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Course Code				Course Title							L	T	P	Credits
BT-CSE-306A				Dynamic Web Application Using PHP							3	-	-	3
											CIE	SEE	Total	
											40	60	100	
Course Outcomes														
CO1	Describe the purpose, history, and setup process of PHP as a server-side scripting language.													
CO2	Develop and apply PHP syntax, operators, control structures, and user-defined functions to create dynamic content.													
CO3	Create and validate HTML forms with PHP using GET and POST methods and store/retrieve data using MySQL.													
CO4	Analyze PHP's use of sessions and cookies to manage and persist in user data across web applications.													
CO5	Evaluate and implement basic file handling capabilities and upload functionalities through PHP-based web solutions.													
CO6	Demonstrate the web security principles to defend PHP applications from common attacks (e.g., SQL Injection, XSS).													
Course Outcomes (CO) to Program Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2										1	2	
CO2	3	2	2		2					1		1	3	2
CO3	3	3	3		3				1	1		1	3	3
CO4	2	2	2		2				1	1		1	2	3
CO5	2	2	3		2						1	1	3	2
CO6	3	3	2		2	3	2	3		1	1	2	3	3
Avg	2.7	2.3	2.4	0.0	2.2	3.0	2.0	3.0	1.0	1.0	1.0	1.2	2.7	2.6

Course Outline

Unit I: Introduction to PHP

Contact Hours: 12

History and Basics- Overview of PHP, its purpose, and how it works in a web development context, Setting up the Environment- Installing and configuring a local development environment (e.g., LAMP stack: Linux, Apache, MySQL, PHP), PHP Syntax- Understanding the basic syntax, including opening and closing tags, comments, and data types (integers, strings, Booleans, arrays, etc.), Variables and Constants - Declaring and using variables, understanding variable scope, and working with constants, Operators- Arithmetic, comparison, logical, and assignment operators, Control Structures - If/else statements, switch statements, loops (for, while, do-while), and break/continue statements.

Unit II: PHP Functions

Contact Hours: 10

User-defined Functions: Creating and calling functions, passing arguments, and returning values. Built-in Functions: Exploring commonly used string functions (strlen, str_replace, etc.), math functions, and other useful PHP functions. Anonymous Functions and Variable Functions: Understanding these advanced function concepts. PHP Arrays: Creating and Accessing Arrays: Indexed and associative arrays, adding and removing elements, and iterating through arrays. Array Functions: Using various array functions for sorting, searching, and manipulating array data. □ Multidimensional Arrays: Working with arrays within arrays.

Unit III: Working with Forms

Contact Hours: 10

HTML Forms: Creating forms using HTML elements (text fields, text areas, radio buttons, checkboxes, dropdowns, etc.), GET and POST Methods: Understanding the differences between GET and POST methods for



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submitting form data. Form Validation: Implementing client-side and server-side validation to ensure data integrity. Working with Multiple Forms: Handling multiple forms on a single page. PHP and Databases (MySQL): Introduction to MySQL: Creating databases, tables, and understanding basic SQL commands. Connecting to MySQL: Establishing a connection between PHP and a MySQL database. □ Executing SQL Queries: Inserting, selecting, updating, and deleting data using PHP. Sessions and Cookies: Introduction to Sessions: Understanding the purpose of sessions and how they work. Starting, Managing, and Destroying Sessions: Registering session variables, accessing session data, and ending sessions. Cookies: Setting, retrieving, and deleting cookies.

Unit IV: File Handling

Contact Hours:10

Reading from and Writing to Files: Opening, reading, and writing data to files. □ File Uploads: Handling file uploads through HTML forms. Security in Web Development: Cross-Site Scripting (XSS) Prevention: Protecting against XSS attacks. SQL Injection Prevention: Using prepared statements and escaping user input to prevent SQL injection. Advanced Topics (Potentially): JAX (Asynchronous JavaScript and XML): Making asynchronous requests to the server without full page reloads. Web Services: Introduction to APIs and consuming web services. Frameworks (e.g., Laravel, Symfony): Brief overview of popular PHP frameworks.

Text Books:

1. "PHP: The Complete Reference" by Steven Holzner: A comprehensive guide to PHP programming.
2. "PHP & MySQL Web Development" by Luke Welling and Laura Thomson: A comprehensive guide covering PHP, MySQL, and web development techniques.

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Course Code				Course Title						L	T	P	Credits	
BT-CSE-380A				Dynamic Web Application Using PHP Lab							-	4	2	
										CIE	SEE	Total		
										50	50	100		
Course Outcomes														
CO1	Apply core PHP syntax, control structures, and functions to create interactive and modular web applications.													
CO2	Design and validate web forms using POST methods, server-side validation, and error handling for secure data processing.													
CO3	Implement CRUD operations and session management using PHP and MySQL to build dynamic web applications.													
CO4	Integrate cookies, file uploads, and security practices to enhance user experience and protect web applications from common vulnerabilities.													
Course Outcomes (CO) to Program Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: high)														
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2		3	-						2	2	2
CO2	3	3	2		3	2				2		2	3	2
CO3	3	3	3	2	3	-				2		3	3	3
CO4	3	3	3	2	3	3				2		3	2	3
Avg	3	2.75	2.5	2	3	2.5	-	-	-	2	-	2.5	2.5	2.5

Lab Experiment:

1. Introduction to PHP Syntax and Control Structures

Statement: Write a PHP script to create a simple calculator that accepts two numbers and an operator (+, -, ×, ÷) through an HTML form, processes the operation using if-else or switch, and displays the result.

2. User-Defined Functions

Statement: Write a PHP script for Grade Evaluation using a user-defined function calculate Grade (\$marks) that accepts marks as input and returns a grade (A/B/C/D/Fail) based on given conditions. Take input through an HTML form and display the grade.

3. Managing Student Data using Arrays

Statement: Create a PHP program to store a list of 5 students using associative arrays. Each student should have a name, roll number, and grade. Display the records in a tabular format using loops.

4. Student Registration Form with POST Method

Statement: Design an HTML form to collect student information (Name, Email, Roll No., Gender, Course). Use the POST method to send this data to a PHP script and display the submitted data.

5. Form Validation and Error Messages

Statement: Modify Program 4 to include server-side validations: check for empty fields, valid email format, and character limits. Display error messages if validation fails.

6. Store Form Data in MySQL Database

Statement: Extend Program 5 to insert the validated form data into a MySQL database named student_portal. Use mysqli or PDO to connect and insert the student record.

7. Display Student Records from Database



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Statement: Write a PHP script to fetch and display all student records from the student_portal database in an HTML table. Highlight even rows using CSS.

8. Edit and Delete Student Records

Statement: Enhance Program 7 by adding "Edit" and "Delete" buttons for each student record. Implement edit functionality to update records and delete functionality to remove them from the database.

9. User Login System using Sessions

Statement: Create a login system where the user (admin/student) can enter a username and password. Authenticate credentials using stored values in the database and use sessions to manage user login status.

10. Cookies for Storing Theme Preferences

Statement: Add functionality where users can select a theme (Light/Dark) using a dropdown. Store this choice in a cookie and use it to set the theme whenever the user revisits the page.

11. Upload Profile Picture

Statement: Extend the registration form to allow students to upload a profile picture (JPG/PNG). Store the image in a server directory and save its path in the database. Display the image along with student details.

12. Secure Student Portal

Statement: Update previous programs to protect against SQL Injection and XSS. Use prepared statements for database queries and sanitize all inputs/outputs.

Final Integration Project

Title: Create a Dynamic Student Portal System

Objective: Combine all the above programs into a single integrated PHP-based portal. The system should support user login, student registration, image upload, CRUD operations, theme preferences using cookies, and security measures.



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Course Code			Course Title								L	T	P	Credits
BT-CSE-403A			Advanced Web Framework								3	-	-	3
											CIE	SEE	Total	
											40	60	100	
Course Outcomes														
CO1	To comprehend the role of web applications framework													
CO2	To develop web applications using framework													
CO3	To apply Laravel framework in web in web applications development													
CO4	To apply Flask framework in web in web applications development													
CO5	To apply advanced framework in deployment													
Course Outcomes (CO) to Program Outcomes (PO) mapping (scale 1: low, 2: Medium, 3: High)														
COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	-	-	-	-	-	-	2	1	-
CO2	3	3	2	1	2	-	-	-	-	-	-	3	2	1
CO3	3	3	2	1	2	-	-	-	-	-	-	3	2	1
CO4	3	3	2	1	2	-	-	-	-	-	-	3	2	1
CO5	3	3	2	1	2	-	-	-	-	-	-	3	2	1
Avg	3	2.8	1.8	1	2	-	-	-	-	-	-	2.8	1.8	1

Course Outline

Unit 1: Web Application Frameworks Concept

Contact Hours: 10

Introduction to Web Application Frameworks, Role of a Framework in Web Development, Types of Web Frameworks, Backend Development with Frameworks, Authentication & Authorization in Frameworks, Error Handling and Debugging

Unit: 2 Full-Stack Development with Web Frameworks

Contact Hours:10

Introduction to Full-Stack Development, Introduction to Node.js, React.js, Vue.js, Full-Stack Development: Integration, Authentication, Sessions, and Security, Deployment, Optimizing performance and scalability

Unit: 3 Laravel Framework

Contact Hours:10

Introduction to Laravel Framework, Laravel's MVC Framework, Eloquent ORM and Database Migration, Authentication and Authorization in Laravel Framework, RESTful APIs with Laravel, Testing of Laravel Framework, Deployment Laravel on Cloud Platforms.

Unit: 4 Flask Framework

Contact Hours:10

Introduction to Flask Framework, Flask Setup and Structure, Building APIs with Flask, Deployment Flask on Cloud Platforms

Textbook:

- Godbole, Achyut, and Atul Kahate. "Web Technologies." McGraw Hill Education (India) Private Limited, 2013.



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2. Minnick, C. Beginning ReactJS foundations building user interfaces with ReactJS: An Approachable Guide, OReilly, 2022.
3. Dean, J., Web Programming with HTML5, CSS, and JavaScript. Jones & Bartlett Learning, 2018.

Reference Books

1. Beaird, Jason, Alex Walker, and James George. The principles of beautiful web design. SitePoint Pty Ltd, 2020.
2. Rebah, H.B., Boukthir, H. and Chedebois, A., Website Design and Development with HTML5 and CSS3. John Wiley & Sons, 2022.

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